Representational Compatibility Is Not Scientific Identity: Coordinate-Governed Mapping of Source-Local Scientific Projections

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Abstract

Scientific knowledge integration can hide the coordinate that determines whether two claims agree, making a single canonical graph overconfident. We introduce epistemic portrait envelopes, the global portraits compatible with source-local observations and licensed mapping constraints. The framework separates point identification, information value and robust action before identification is complete. On a prospectively frozen 32-case public-reference holdout, coordinate-governed mapping made no false merges, versus 0.1875 for flat predicate canonicalization (paired difference -0.1875, 95% bootstrap interval $[-0.34375, -0.0625]$). A constructed 36-case corpus gives every deterministic observation-only rule an exact 27/36 ceiling. On one historical OAEI 2004 test-103 case, BERTMap recovered 33/33 common class pairs versus AML's 8/33; full-task F1 was 33/62 versus 16/137, with 58 property correspondences unmatched. This comparison is descriptive, not population evidence or general superiority. The results support scoped integration under partial identification; naturalistic validity and downstream gains remain prospective.

Keywords: scientific knowledge integration; scientific identity; provenance; schema alignment; measurement harmonization; scientific pluralism; knowledge representation.

1 Introduction

Scientific claims are distributed across disciplines, each with its own terminology, measurement practices, contexts, and representational choices. A global synthesis can therefore fail even when every local statement is individually well formed: identical surface terms can refer to different objects, different labels can denote compatible objects, and apparently contradictory claims can differ only in modality, attribution, or context. The central problem of this paper is not retrieval itself, but the authority to identify two already-structured scientific representations as one global statement.

1.1 The proposal in brief

Each source is represented as a *source-local scientific projection* bound to its provenance. A proposed cross-source mapping records scientific coordinates such as referent, construct, measurement, context, polarity, modality, and attribution separately rather than reducing them to one similarity score. When the coordinates required by the claim are compatible, the projections may be *glued* into a single global statement. When a load-bearing coordinate is incompatible or unresolved, the mapping remains a typed *obstruction* or a *plural view* rather than being forced into a canonical merge.

That rule is evaluated at the layer for which external evidence currently exists. An initial public-reference study contains 32 already-structured cases derived from pinned external sources. A second, disjoint 32-case holdout was selected and execution-frozen before its system outputs, including its exact gold digest, source revisions, evaluator code, bootstrap seed, margins, and pass rule. On that confirmatory holdout, coordinate-governed mapping produced zero false merges against 0.1875 for flat predicate canonicalization; the paired difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. The false-split difference against an exact-coordinate conservative control was 0.000 with interval $[0.000, 0.000]$, satisfying the rule declared before those outputs were seen.

The evidence is deliberately narrower than the mechanism's design ambition. It does not establish raw-text extraction superiority, expert construct validity across the eight case families the broader design describes, reader-level recoverability of a generated portrait, downstream scientific-answer improvement, or necessity of every semantic coordinate. Each of those remains undetermined rather than being inferred from the mapping result.

The surviving empirical claim is the narrow one: for the frozen public-reference mapping task, explicit scientific-coordinate conditions plus an obstruction state reduce false merges relative to flat predicate canonicalization while satisfying the false-split guard declared in advance. The initial 32 cases are not pooled into the confirmatory verdict, and coordinate ablations with no measured effect are reported as coverage limits rather than as proof that those coordinates are unnecessary.

1.2 Structure of the paper

Section 2 positions the mapping rule among the extraction, schema-alignment, integration, stance and pluralism literatures it builds on. Section 3 defines source projections, mappings, gluing, obstruction, and the three-valued reporting discipline. Section 5 describes the two public-reference mapping datasets and their provenance. Section 6 records the prospective confirmatory design and its controls. Section 9 reports the frozen result and the ablations. Section 10 separates the supported mapping claim from what remains unmeasured, Section 11 states the bounded conclusion, and Section 12 lists every artifact behind a reported number.

2 Related work

The question asked here sits downstream of most of the machinery that cross-domain scientific integration usually needs. Extraction, schema matching, entity resolution, fusion, contradiction and stance analysis, and literature-based discovery are all treated as available capability. What remains is smaller and, we argue, unresolved by any of them: given two already-structured source projections, which coordinate differences license a merge, and which oblige the system to keep a provenance-bound obstruction visible instead.

2.1 Structured scientific knowledge bases and extraction

MUSE [8] constructs a full-text, multi-domain resource of source-grounded problem–solution–rationale structures, and the Discovery Engine [2] distils literature into structured knowledge artifacts and an integrated computational landscape. Both target the representation itself: what a source contains, recovered at scale. The decision studied here begins after that representation exists, and asks whether referent, construct, measurement, context, modality and attribution distinctions can govern a downstream mapping outcome and produce an obstruction when compatibility has not been earned. SciER [30] supplies scientific entity and relation annotations for datasets, methods and tasks in full text, which makes information extraction, discourse processing and entity/relation recovery an input to this work rather than part of it.

2.2 Schema-centric approaches

SciSchema.org [10] provides multidisciplinary, expert-reviewed scientific-process schemas with parameters, measurements, procedures and provenance, and is strong prior art for reusable scientific schema design. SCOPE and SCION [14] introduce an auditable benchmark and reference pipeline for evidence-linked schema induction and optional fusion from text, including conservative fusion behaviour. Executable Schema Contracts [16] discover an executable schema from heterogeneous sources and use it to drive provenance-aware construction and multi-source retrieval. Each of these produces a shared structure that a downstream consumer can rely on; none of them is asked whether a particular proposed mapping preserves the coordinates a specific scientific claim depends on. That is the discriminator evaluated here, over already-structured projections and including cases in which refusing a merge is the correct outcome.

2.3 Alignment, integration, and resolution

Ontology matching [11] studies correspondences such as equivalence, subsumption and disjointness between ontology entities, and language-model systems such as LLMATCH [29] add modern schema-preparation, candidate-selection and column-alignment machinery. Surveys of knowledge-graph construction [13] cover heterogeneous-source integration, ontology and schema matching, entity resolution, fusion, provenance and quality assurance, while reviews of scholarly knowledge graphs [24] cover extraction, construction, refinement and use of semantic scholarly infrastructures. The distinction pursued here is that a mapping decision is bound to several scientific coordinates at once, and that the exact source, provenance and non-preserved-coordinate information is retained with the resulting decision. Entity resolution and fusion are especially useful pressure cases, because a record-level match may be appropriate at exactly the moment a claim-level merge remains unsafe on construct, measurement, context or modality grounds.

2.4 Claim comparison and stance

Stance detection assigns target-relative labels such as support or opposition; surveys document its use in fact-checking and misinformation settings [12]. Apparent scientific disagreement, however, frequently arises from different context, modality, measurement or attribution coordinates rather than from a genuine opposition, and a mapping record that collapses every difference into one generic contradiction label discards the reason. Preserving that reason, rather than detecting stance, is what is evaluated here.

2.5 Scientific-variable semantics and interoperability

The I-ADOPT interoperability framework [19] provides a community-developed semantic decomposition for machine-readable observable and scientific-variable descriptions, and a recent benchmark builds an expert-annotated corpus of more than one hundred scientific variables and reports that large language models still struggle to construct those structured representations accurately [18]. FAIR 2.0 separates terminological, ontological, referential, schema and logical interoperability, and argues that semantic interoperability requires explicit mappings rather than one privileged ontology [26]. Recent provenance work goes further: DEC reads provenance-bearing statements as epistemic stance and preserves source-relative disagreement instead of collapsing every attributed assertion into a single factual core [25]. Structured variable semantics, generic semantic interoperability and provenance-aware pluralism are therefore supplied by this literature. The residual question is claim-relative and downstream of all of it: once a strong semantic integration stack has produced structured candidate mappings, what further conditions license an assertion of *scientific identity*? The decision contract studied here consumes ontology and schema alignment, provenance, plural-view machinery and consistency checks as inputs, and returns a glue, obstruction, plural-view or unresolved outcome from load-bearing construct, measurement, context, provenance-stance and global-composition conditions.

2.6 Measurement equivalence and construct validity

Measurement-equivalence and replication literatures established long ago that nominally identical outcome measures need not be comparable across settings, and that equivalence can require explicit testing [15]. The mechanism here makes measurement and operationalization one explicit coordinate of the mapping record and allows an unresolved measurement relation to block a global merge on its own.

2.7 Literature-based discovery and pluralism

Swanson's literature-based discovery lineage [21, 22] shows how complementary literatures can expose implicit connections without direct citation links. The bounded requirement added here is on the bridge concept itself: it must retain compatible scientific meaning across the two source-local projections, or the bridge stays an obstruction rather than being promoted to a scientific integration. Scientific pluralism has similarly argued that multiple representations may remain legitimate without collapsing into one complete account [17, 7]; the computational question is whether an integration record can carry a plural or obstructed outcome, with exact source lineage, instead of forcing a canonical merged statement.

2.8 Scientific retrieval and cross-domain synthesis

OpenScholar [1] performs retrieval-augmented synthesis over the scientific literature at scale, and BioSage [27] combines retrieval, specialized agents, cross-disciplinary translation and reasoning for scientific tasks. Retrieval, synthesis, agent orchestration and cross-domain translation are baseline capability for the problem, as are language-model-assisted schema matching and induction. The experiment reported here deliberately holds all of that fixed and varies only the decision rule applied after scientific projections are already structured, so that the construct-validity question can be asked without a retrieval confound.

The empirical claim is correspondingly narrow. It is not that this combination solves raw-text scientific integration end to end; it is that on a disjoint, prospectively frozen public-reference holdout of already-structured scientific projections, the typed mapping calculus reduced false merges against flat predicate canonicalization while satisfying the false-split guard declared in advance. Raw-text extraction, full expert construct validity, recoverability and downstream utility are separate questions with separate evidence requirements.

3 Method

This section defines the scientific mapping objects the public-reference evaluation is defined over. The definitions are data-level: they do not depend on a particular language model or retrieval backend.

The implementation evaluated here is the ORION knowledge-integration layer; it is the artifact under test, and the rest of the paper describes the mapping rule rather than the product, so that each claim reads as a claim about the rule. Its source, protocols and evidence archives are listed in Section 12.

3.1 Three-valued outcomes

Every mapping decision measured in this paper is three-valued rather than binary. A proposed mapping may be determined admissible, determined inadmissible, or left *unresolved*; a coordinate relation may be determined compatible, determined incompatible, or left unresolved. An outcome that could not be determined is recorded and counted separately from an outcome determined to be false, and the two are never merged. The distinction is load-bearing rather than decorative: missing source authority, an absent coordinate or an unsupported relation produces an unresolved outcome, and an unresolved outcome is not evidence that the mapping is wrong. Throughout the paper, an outcome described as undetermined is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

3.2 Source and source-local projection

Let a source record s bind an external document or structured source item to an exact identity, version or revision, locator, and content commitment. The relevant scientific meaning is represented as a source-local projection

$$P_s = \langle R_s, C_s, M_s, X_s, \rho_s, \mu_s, a_s, \delta_s, \Pi_s \rangle, \tag{1}$$

with coordinates for referent R_s, construct C_s, measurement/operationalization M_s, context X_s, polarity ρ_s, modality μ_s, attribution a_s, discourse relation δ_s, and assumptions/unresolved information Π_s.

Each coordinate is a typed hypothesis supported by the source authority available for that case, not a value filled merely to complete a schema. In the public-reference datasets, unsupported coordinates remain absent or unresolved; the study does not ask a model to infer missing gold coordinates from raw papers.

3.3 Coordinate-wise comparison

A comparison between projections P_a and P_b is a vector of typed coordinate relations rather than a scalar similarity score:

$$\begin{aligned} q(P_a, P_b) = \langle &\mathrm{ref}(R_a, R_b), \mathrm{const}(C_a, C_b), \mathrm{meas}(M_a, M_b), \mathrm{ctx}(X_a, X_b), \\ &\mathrm{pol}(\rho_a, \rho_b), \mathrm{mod}(\mu_a, \mu_b), \mathrm{attr}(a_a, a_b), \mathrm{disc}(\delta_a, \delta_b) \rangle. \end{aligned} \tag{2}$$

Relations can express compatibility, incompatibility, conditional transformability, or unresolved evidence according to the frozen semantic taxonomies. A proposed relation does not acquire authority merely from lexical or embedding similarity.

3.4 Typed mapping and preservation conditions

A typed mapping m states which coordinate relations it licenses and under what preservation conditions. Its record contains:

- the proposed relation (for example identity, equivalence, conditional transform, subsumption, association-only, no-licensed-mapping, or unresolved);
- the coordinates the mapping claims to preserve;
- conditions that must hold for the mapping to be used; and
- coordinates or information that are not preserved and therefore may not be silently transferred into a global statement.

This representation is deliberately stricter than a successful match score: a mapping can be structurally plausible and still be inadmissible for the scientific claim if a load-bearing condition is absent.

3.5 Gluing, obstruction, and unresolved mapping

Two or more projections may be *glued* into one global statement only when the coordinate relations the mapping requires are compatible and its preservation conditions are satisfied. If a required relation is incompatible, the correct output is a typed *obstruction* naming the failing coordinate or condition. If the available evidence is insufficient, the relation stays unresolved rather than being forced to merge or split. Multiple locally valid but mutually incompatible views may remain a *plural view*.

The confirmatory experiment tests this decision semantics directly. Its strongest discrimination is on polarity, modality, attribution and context cases, where flat predicate canonicalization merges claims that the typed rule keeps separate. It does not identify the necessity of every coordinate; coordinate necessity requires targeted cases that the current datasets do not contain.

3.6 Provenance and recoverable records

Every case retains the external source identity and the frozen record used to derive the expected mapping decision. A mapping result retains the source records and the coordinate conditions supporting its terminal. That provenance is what makes the reported result independently replayable.

The wider design goes further and defines a global portrait

$$\mathcal{G} = \langle \{P_i\}, \mathcal{M}, \sigma \rangle, \tag{3}$$

where $\mathcal{M}$ stores accepted and rejected mappings and σ maps a global statement back to source projections and conditions. Whether a reader can in practice recover a generated portrait this way is a separate empirical question and is not answered by the current mapping study.

3.7 Search-universe feedback is outside the evaluated claim

The wider design also allows newly absorbed representations to change the search universe W and reopen future acquisition routes. That feedback is not manipulated or measured in the public-reference mapping experiment. No result in this paper is attributed to a "W effect," and no expansion of W is used as an ablation supporting the reported mapping result.

3.8 What is and is not claimed

Nothing here claims the first scientific knowledge graph, cross-domain retrieval-augmented system, schema matcher, provenance graph, entity resolver or literature-discovery method. The supported empirical object is narrower: over the frozen public-reference mapping cases, explicit scientific-coordinate conditions plus an obstruction state reduce false merges relative to flat predicate canonicalization while satisfying a conservative false-split guard.

Local exact worlds continue to falsify implementation semantics in controlled cases, in the manner established for source-local reconstruction and for typed contradiction detection [4, 5]. They do not establish external raw-text extraction quality, recoverability, downstream utility, or broad cross-domain generality; those remain separate prospective questions.

3.9 Method-structure projection

Scientific meaning coordinates describe what a source claims. Structural learning downstream of that also needs a source-local account of *how the source's method transforms its problem*, so the representation is extended with a versioned method-structure projection. It is derived from a provenance-bound method-realization record of the kind produced by source-local epistemic reconstruction [4], and retains exact source, document, version and span identity, the target role and initial obstruction, assumptions and preconditions, representation changes and auxiliary objects where the source supports them, the transformation graph, invariants, progress and termination semantics, reconstruction to the original target, failure and negative-result statements, explicit author rationale when the source actually supplies it, and coordinates typed as unknown otherwise.

This projection stays source-local. In particular, the absence of an author rationale is not permission to invent one. The projection stores the exact realization and span digests that licensed each structured interpretation, and carries no authority to declare that two methods belong to the same formal family.

3.10 Cross-domain method alignment and plural views

Given two source-local projections, a claim-relative mapping μ may be proposed by comparing declared coordinates such as preconditions, assumptions, invariants, progress, termination and reconstruction. The alignment outcome is one of aligned, related, obstructed, or unresolved. The record preserves both projection identities, source-local names and meanings, unmatched assumptions, coordinate-level support and obstruction, and the exact reasons for refusing a merge. An aligned outcome means only that the declared source-local coordinates agree for the proposed mapping purpose; establishing formal equivalence is outside this mechanism's authority.

A clean vocabulary-changing example is numerical bisection against monotone threshold calibration. Their operation names differ, but under the narrow purpose "bounded monotone localization" both require an ordered domain and a bracketed monotone target, preserve a target-containing bracket, make progress by shrinking the admissible interval, terminate at a tolerance, and reconstruct a representative in the original target space. A candidate alignment may therefore be recorded while each donor's vocabulary and lineage is preserved.

A false-merge control uses the same-looking midpoint-and-discard-half surface sequence on a non-monotone response. The monotonicity and bracketing contract no longer holds and the invariant can fail, so the correct result is an obstruction; topology or lexical similarity cannot override that coordinate. A third control uses an opaque recovery procedure whose source does not establish progress or reconstruction semantics. Even when its known effects resemble a rollback method, the alignment stays unresolved rather than filling the missing coordinates from analogy.

Plural views are therefore first-class outcomes: structurally suggestive sources need not collapse into one global method. An obstruction certificate records exactly which coordinates prevent a merge and keeps both source projections recoverable.

3.11 Downstream structural-learning boundary

What this mechanism owns is source-local method projection and candidate plural alignment. Claim-relative structural reduction, substitution and method-family membership are outside it, as is any learned distribution over these versioned objects and any generated method structure proposed from such a distribution. Neither learned similarity nor generated structure changes a source projection or confers scientific authority on it. The bounded pilot reported in Section 6.7 tests only representation and alignment non-vacuity; it is not evidence of general extraction from arbitrary papers.

4 Epistemic portrait envelope theory

The mapping calculus above can be stated more generally. Scientific sources rarely reveal a unique global state. They reveal projections of that state, with different coordinates observed, suppressed, coarsened, or measured under different operationalizations. The appropriate integration object is therefore not always one completed graph. It is an envelope of the global portraits still compatible with the observations and their source-bound constraints. This section develops that object and separates three questions: what the sources identify, what additional information could identify, and what a downstream decision may safely do before identification is complete.

The construction imports the logic of partial identification [20, 23], comparison of information [6], and robust decision theory [3, 28]. The contribution is not a new claim to those donor theories. It is their claim-relative composition with provenance-bound scientific projections, typed mapping constraints, and global-portrait construction.

4.1 Worlds, observation fibres, and portrait envelopes

Let Ω be a set of scientifically admissible worlds. A world $\omega \in \Omega$ fixes the source records, the source-to-projection relations, and a global portrait $G(\omega) \in \mathcal{G}$. The observation operator

$$O : \Omega \longrightarrow \mathcal{Y} \tag{4}$$

returns exactly the evidence available to the integration procedure: the observed projection coordinates, explicit missingness, whatever source identity/version information the interface exposes, and the mapping constraints licensed by those sources. It does not fill an unobserved coordinate from the world that generated it.

The operator is interface-relative. If a procedure receives source text and semantic provenance, those are part of O; if it receives only projections, they are not. Opaque record identifiers that carry no licensed scientific meaning are quotiented by bijective relabelling rather than treated as a lookup key for memorising gold. Every impossibility statement must name this observation boundary.

Definition 1 (Observation fibre and epistemic portrait envelope)**.** For observed evidence $y \in \mathcal{Y}$, define

$$\Omega_y = \{\omega \in \Omega : O(\omega) = y\}, \qquad \mathfrak{E}(y) = \{G(\omega) : \omega \in \Omega_y\}. \tag{5}$$

Ω_y is the observation fibre and $\mathfrak{E}(y)$ is the epistemic portrait envelope. A query $q : \mathcal{G} \to \Theta$ has identified set

$$\Theta_q(y) = \{q(g) : g \in \mathfrak{E}(y)\}. \tag{6}$$

The query is point identified at y exactly when $\Theta_q(y)$ is a singleton.

Write $\mathcal{Y}_\mathrm{r} = O(\Omega)$ for the set of realised observations. If $y \notin \mathcal{Y}_\mathrm{r}$, then Ω_y and $\Theta_q(y)$ are empty. We treat such an input as an explicit INVALID MODEL/INTERFACE terminal, denoted $\perp_\mathrm{inv}$, rather than as an ordinary scientific envelope. In particular, no robust loss is optimized by taking a supremum over the empty set. All decision statements below are restricted to $y \in \mathcal{Y}_\mathrm{r}$ unless an explicit invalid terminal is mentioned.

The fibre calculus is set-theoretic and pointwise. A globally measurable decoder is an additional conclusion, not a consequence of fibre constancy on arbitrary measurable spaces. If $\Omega, \mathcal{Y}$, and the query codomain carry sigma-algebras, such a conclusion requires a measurable factorisation $q \circ G = d \circ O$. Where conditional probabilities are invoked below, they are separately licensed probabilistic objects. If derived from a joint law, we assume standard-Borel spaces and a declared version of the regular conditional law. The set-valued portrait envelope alone does not manufacture a conditional probability.

The query can ask whether two projections may be glued, which typed obstruction holds, what numerical estimand a merged record implies, or which downstream action is warranted. Identification is therefore claim-relative: an envelope can identify one query while leaving another unresolved. A portrait need not be completed in every coordinate before it licenses a specific conclusion.

The envelope also separates scientific plurality from epistemic uncertainty. Plurality is a feature represented *inside* a feasible portrait: several incompatible but source-valid views coexist. Epistemic uncertainty is variation *across* feasible portraits because the available observation does not decide which completion obtains. A plural-view terminal can therefore be point identified; an unresolved terminal cannot.

4.2 The fibre criterion and an impossibility result

Theorem 1 (Fibre criterion for scientific identification)**.** *For any observation y with $\Omega_y \neq \varnothing$ and query q, the following are equivalent:*

1. *q is point identified at y;*
2. *$q \circ G$ is constant on Ω_y;*
3. *there exists an observation-only answer function $d(y)$ that equals $q(G(\omega))$ for every $\omega \in \Omega_y$.*

If two worlds in the same observation fibre have different query values, no deterministic procedure reading only y can be correct in both worlds.

Proof. The image of Ω_y under $q \circ G$ is exactly $\Theta_q(y)$. It is a singleton if and only if the map is constant on the fibre. In that case its unique value defines $d(y)$. Conversely, if $d(y)$ is correct for every world in the fibre, all those worlds must have query value $d(y)$. The final claim follows immediately: the procedure receives the same input in both worlds and therefore emits the same answer, which cannot equal two different values. □

This theorem turns a partial-observation failure into a scientific result rather than an invitation to invent a fifth rule over the same information. When a coordinate erased by extraction is the only fact distinguishing two source relations, model capacity, prompt changes, and a more elaborate comparison rule cannot restore it. The remedy must either acquire information, declare an assumption that removes worlds from the fibre, or return a set-valued/unresolved answer.

4.3 A canonical sufficient interface

The envelope is not a demand that every integration system share one internal architecture. It defines the least claim-relative interface a downstream consumer must be able to recover.

Definition 2 (Envelope sufficiency)**.** A summary $S : \mathcal{Y} \to \mathcal{Z}$ is sufficient for query q when there exists a decoder h such that

$$\Theta_q(y) = h(S(y)) \qquad \text{for every } y \in \mathcal{Y}_\mathrm{r}. \tag{7}$$

Thus equal summaries may never hide different identified sets.

Theorem 2 (Universal factorisation of sufficient interfaces)**.** *On $\mathcal{Y}_\mathrm{r}$, the map $T_q : y \mapsto \Theta_q(y)$ is sufficient for q. Every other sufficient summary S refines it: there is a map h with $T_q = h \circ S$. Consequently, all sufficient implementations evaluated only through the same set-robust loss have the same robust loss table and the same set of minimax actions, up to ties.*

Proof. T_q is decoded by the identity map on realised observations. The factorisation for another sufficient S is its defining decoder. Robust upper loss for action a is a function only of the recovered set, $\sup_{\theta \in T_q(y)} L(a, \theta)$. Equal recovered sets therefore give the same value for every action, and minimisation gives the same optimal-action set. □

This theorem explains both the ambition and the limit of the architecture. A semantic product that exposes exactly the same identified set is not an inferior baseline; it is an information-equivalent implementation and must tie at this interface. A product that collapses two observations with different identified sets is insufficient for that query, no matter how rich its remaining representation is. Any empirical advantage must therefore come from a genuinely finer observation, justified constraints, a more faithful approximation to the envelope, or a different declared downstream loss—not from the name or central location of the layer. The conclusion is deliberately about the displayed set-robust rule. Two summaries that recover the same identified set may still support different Bayes decisions if one additionally carries a licensed probability law over that set.

4.4 Information refinement and the observation floor

Say that O_2 refines O_1 when there is a map r such that $O_1 = r \circ O_2$. Thus O_2 retains at least as much information as O_1; the relation is deterministic and makes no probabilistic sufficiency claim. Let $\mathcal{A}$ be a finite nonempty downstream action set and $L : \mathcal{A} \times \Theta \to [0, \infty)$ a declared loss. Define the robust decision floor, for $y \in \mathcal{Y}_\mathrm{r}$,

$$B_q(y) = \min_{a \in \mathcal{A}} \sup_{\theta \in \Theta_q(y)} L(a, \theta). \tag{8}$$

Theorem 3 (Information-refinement monotonicity)**.** *If O_2 refines O_1, then for every realised world ω,*

$$\Theta_q^{(2)}(O_2(\omega)) \subseteq \Theta_q^{(1)}(O_1(\omega)) \quad \textit{and} \quad B_q^{(2)}(O_2(\omega)) \leq B_q^{(1)}(O_1(\omega)). \tag{9}$$

Proof. Every world having the refined observation $O_2(\omega)$ also has coarse observation $r(O_2(\omega)) = O_1(\omega)$, so the refined fibre is a subset of the coarse fibre. Taking images under $q \circ G$ preserves inclusion. For any fixed action, the supremum of its loss over a subset cannot increase; taking the minimum over actions preserves the inequality. □

The quantity $B_q(y)$ is an observation floor, not a defect score for a particular algorithm. It can fall when truthful extraction, measurement, provenance, or source acquisition refines the observation. It cannot be made to disappear merely by changing the decision rule over the same fibre. For a candidate rule δ, its floor-adjusted avoidable harm is

$$X_\delta(y) = \sup_{\theta \in \Theta_q(y)} L(\delta(y), \theta) - B_q(y) \geq 0. \tag{10}$$

This is the quantity a successor experiment should compare. Counting total error without subtracting or stratifying by the observation floor confounds avoidable decision failure with information that no compared rule received.

4.5 Coordinate opportunity is within-comparison support

The same fibre logic applies to experimental support. Suppose a protected evaluation is organised into comparison blocks $i = 1, \ldots, n$, with two source-grounded members y_i^L, y_i^R in each block. For scientific coordinate c, let $z_c(y)$ be its candidate-visible value and define the coordinate-opportunity matrix

$$S_{ic} = \mathbf{1}\left\{ z_c(y_i^L) \neq z_c(y_i^R) \right\}. \tag{11}$$

This is a support condition, not an outcome. A coordinate may take many different values over the atlas while its entire column of S is zero: the value changes between blocks but never supplies a comparison inside one.

Theorem 4 (No coordinate attribution without comparison support)**.** *Fix a coordinate c and a paired design $\mathcal{D} = \{(y_i^L, y_i^R)\}_{i=1}^n$. If $S_{ic} = 0$ for every i, then any two decision rules that agree on the observed design support are observationally indistinguishable. In particular, when the admissible domain contains an off-support pair with a c-contrast, the design cannot distinguish a rule that is insensitive to c from a rule that differs only on such pairs. Consequently, marginal variation of z_c across blocks does not identify coordinate-specific decision value, harm reduction, or necessity without an additional cross-block exchangeability or structural extrapolation assumption.*

Proof. The first claim follows because a statistic of the observed blocks depends only on rule values on that support. For the particular construction, let δ_0 be any rule and let δ_1 equal it on every observed pair but differ on one admissible off-support pair with a c-contrast. The hypothesis $S_{ic} = 0$ ensures that this difference is never queried by $\mathcal{D}$, so the two rules induce exactly the same outputs and losses on every observed block. No statistic of those blocks can distinguish them. Attributing a difference to c from between-block variation therefore requires an assumption linking otherwise different blocks; that assumption is not supplied by the paired design itself. □

The theorem separates three increasingly strong gates. A field being populated is weaker than marginal variation; marginal variation is weaker than nonzero within-block contrast; and within-block contrast is weaker than coordinate-specific identification when several coordinates move together. For a claim about coordinate c, the protocol must therefore report the column sum $\sum_i S_{ic}$, the joint pattern of the other coordinate columns, and the protected query or loss opportunity. A zero column is an automatic CANNOT CHECK for that coordinate. A nonzero column admits analysis but does not by itself prove a causal coordinate effect.

4.6 Credal risk envelopes for merge, split, and unresolved actions

For a realised observation y at which probabilistic authority is available, let the query space be measurable, let $\Theta_q(y)$ be measurable, and let $\mathcal{K}_y$ be a nonempty set of licensed probability measures satisfying $P(\Theta_q(y)) = 1$ for every $P \in \mathcal{K}_y$. For each $a \in \mathcal{A}$, assume $L(a, \cdot)$ is measurable and define its lower and upper expected risks

$$\underline{R}(a \mid y) = \inf_{P \in \mathcal{K}_y} \int L(a, \theta)\, dP(\theta), \qquad \overline{R}(a \mid y) = \sup_{P \in \mathcal{K}_y} \int L(a, \theta)\, dP(\theta). \tag{12}$$

These values may be extended-real for a general nonnegative loss. A robust rule minimizes $\overline{R}(a \mid y)$. Because $\mathcal{A}$ is finite and nonempty, the pointwise minimum is attained. A globally measurable robust rule additionally requires the upper-risk functions to be measurable; with a fixed ordering of $\mathcal{A}$, measurable upper risks give a measurable tie-broken selector.

This credal rule contains the earlier set-robust floor as a special case. If $\mathcal{K}_y$ is the class of all probability measures supported on $\Theta_q(y)$, its upper expected risk equals $\sup_{\theta \in \Theta_q(y)} L(a, \theta)$ because all point masses are available. More generally, replacing $\mathcal{K}_y$ by a nonempty subset can only decrease every upper risk and the resulting minimax floor. This is a refinement of licensed probabilistic authority, distinct from refining the observation operator itself.

For a binary mapping query, let $\theta = 1$ mean that glue is licensed and $\theta = 0$ mean that it is not. The downstream actions are merge M, keep separate S, and unresolved U. A false merge costs $c_{\mathrm{FM}} \geq 0$, a false split costs $c_{\mathrm{FS}} \geq 0$, and deferral costs $c_U \geq 0$. Because a binary law is determined by $p = P(\theta = 1)$, write

$$K_y = \{P(\theta = 1) : P \in \mathcal{K}_y\} \subseteq [0, 1], \qquad \ell_y = \inf K_y, \quad u_y = \sup K_y. \tag{13}$$

The set K_y need not be an interval, convex, closed, or endpoint-attaining.

Proposition 1 (Binary robust action over an arbitrary credal set)**.** *The closed interval hulls determined by the lower and upper risks are*

$$[\underline{R}(M \mid y), \overline{R}(M \mid y)] = [c_{\mathrm{FM}}(1 - u_y), c_{\mathrm{FM}}(1 - \ell_y)], \tag{14}$$
$$[\underline{R}(S \mid y), \overline{R}(S \mid y)] = [c_{\mathrm{FS}}\ell_y, c_{\mathrm{FS}}u_y], \tag{15}$$
$$[\underline{R}(U \mid y), \overline{R}(U \mid y)] = [c_U, c_U]. \tag{16}$$

In particular, the worst-case expected losses are

$$\overline{R}(M \mid y) = c_{\mathrm{FM}}(1 - \ell_y), \quad \overline{R}(S \mid y) = c_{\mathrm{FS}}u_y, \quad \overline{R}(U \mid y) = c_U. \tag{17}$$

Consequently, unresolved is minimax-optimal exactly when

$$c_U \leq \min\{c_{\mathrm{FM}}(1 - \ell_y), c_{\mathrm{FS}}u_y\}. \tag{18}$$

Proof. Merging has risk $c_{\mathrm{FM}}(1 - p)$, a decreasing affine function of p. Its infimum and supremum over K_y are therefore its limiting values at u_y and ℓ_y, respectively, whether or not those endpoints are attained. Keeping separate has risk $c_{\mathrm{FS}}p$, so the order is reversed. Deferral has constant risk. Comparing the three upper risks gives the exact criterion. $\square$

Corollary 1 (Sharp interval and vacuous special cases)**.** *If the licensed credal set is exactly the full sharp interval $K_y = [\underline{p}(y), \overline{p}(y)]$, then*

$$\overline{R}(M \mid y) = c_{\mathrm{FM}}(1 - \underline{p}), \qquad \overline{R}(S \mid y) = c_{\mathrm{FS}}\overline{p}, \tag{19}$$

and Equation (18) *holds with* $(\ell_y, u_y) = (\underline{p}, \overline{p})$. *With no probabilistic authority beyond the vacuous set* $K_y = [0,1]$ *when both binary states remain feasible, unresolved is minimax-optimal exactly when* $c_U \le \min\{c_{\mathrm{FM}}, c_{\mathrm{FS}}\}$.

If $[\underline{p}, \overline{p}]$ is merely an outer bound satisfying $K_y \subseteq [\underline{p}, \overline{p}]$, then $c_{\mathrm{FM}}(1-\underline{p})$ and $c_{\mathrm{FS}}\overline{p}$ are conservative upper bounds on the two robust risks. Optimizing against the whole outer interval is a defensible, more conservative decision problem, and the interval criterion is exact for that enlarged problem. It does *not* by itself certify that unresolved is minimax for the unknown smaller K_y: the alternative actions' true robust risks may be strictly below their outer upper bounds. Exact minimax claims therefore require the actual extrema (ℓ_y, u_y) or a separately justified sharp interval. For example, with $K_y = \{1/2\}$, outer interval $[0,1]$, $c_{\mathrm{FM}} = c_{\mathrm{FS}} = 1$, and $c_U = 3/4$, the outer problem selects unresolved, whereas both determinate actions have true robust risk $1/2$.

The costs are downstream and use-specific. A clinical synthesis may assign a larger cost to a false merge than a discovery interface; a screening system may assign a larger cost to a false split. The portrait mechanism should expose the identified set and provenance needed to evaluate those choices, not silently encode one universal utility function.

4.7 Global composition is a joint-completion problem

Let each accepted source or mapping record impose a set $C_j \subseteq \Omega$ of worlds consistent with its licensed conditions. The jointly feasible set is

$$F(y) = O^{-1}(y) \cap \bigcap_{j=1}^{m} C_j. \tag{20}$$

A global glue requires $F(y) \neq \varnothing$; a claim additionally requires its query to be constant on $F(y)$. Pairwise mapping success does not imply either condition.

Proposition 2 (Pairwise compatibility is insufficient). *There exist three mapping constraints that are pairwise jointly satisfiable but have no common global completion. Hence a system that accepts every pairwise-compatible edge can license an inconsistent global portrait.*

Proof. Take $\Omega = \{1,2,3\}$ and $C_1 = \{1,2\}$, $C_2 = \{2,3\}$, $C_3 = \{1,3\}$. Every pairwise intersection is nonempty, but $C_1 \cap C_2 \cap C_3 = \varnothing$. □

Constraint processing and acyclic join theory provide stronger algorithms for deciding global consistency than this elementary counterexample [9]. The point here is architectural: pairwise ontology or schema correspondences are proposals for C_j, not self-executing authority for a global merge. Empty joint completion is an identified global obstruction. Multiple completions with different query values are partial identification. Multiple source-valid views inside every completion are an identified plural portrait.

4.8 Positive-reference absence without closure authority

Many discovery-style evaluations publish a set of asserted positives rather than a complete binary labelling of a candidate universe. Let U be a frozen candidate universe and let $R^+ \subseteq U$ be the observed positive reference. A binary completion is a map $t : U \to \{\mathsf{glue}, \mathsf{obstruction}\}$ satisfying $t(x) = \mathsf{glue}$ for every $x \in R^+$. The licensed completion class may also contain source-grounded negative assertions or an authority constraint certifying that R^+ is exhaustive. Absent such a constraint, omission is an observation, not a negative label.

Theorem 5 (Positive-reference non-identification without closure authority)**.** *Fix $x \in U \setminus R^+$. Suppose the licensed completion class retains both (i) a completion in which $t(x) = \mathsf{glue}$ and the positive reference omitted that true equivalence and (ii) a completion in which $t(x) = \mathsf{obstruction}$. Then the binary truth at x is not identified by (U, R^+): its identified set is*

$$I_{R^+}(x) = \{\mathsf{glue}, \mathsf{obstruction}\}. \tag{21}$$

Consequently, reference absence alone cannot license an obstruction label or an unconditional point-valued harm score. Assigning obstruction to every $x \notin R^+$ adds a closed-world exhaustivity assumption.

Proof. The two licensed completions induce the same observed pair (U, R^+) but give different binary answers at x. The fibre criterion therefore makes both answers members of the identified set and rules out any uniformly correct singleton decoder based only on (U, R^+). Declaring the second answer unique removes the first completion; that removal is exactly additional negative or exhaustivity authority, not a consequence of reference omission. □

The premise is deliberately authority-relative. The theorem does not assert that every reference alignment is incomplete, nor does a missing licence file prove incompleteness. It states what follows when the admissible-world model has not licensed closure and therefore retains both completions. Explicit disjointness evidence, source-certified negatives, logical constraints, or an authoritative exhaustivity certificate over the frozen universe may remove one completion. Matcher nonselection and agreement among correlated algorithms do not do so by themselves.

This result is not specific to ontology matching. It applies whenever a benchmark records discoveries without certifying the full negative complement. In that setting the appropriate target is an authority-aware identification envelope: point evaluation may be reported only for identified cells, while unlicensed omissions remain set-valued or CannotCheck.

4.9 Direct-certificate calibration

Let $C_G, C_O \subseteq U$ be licensed direct-certificate sets for glue and obstruction. Consider the completion class of all binary maps $t : U \to \{\mathsf{glue}, \mathsf{obstruction}\}$ satisfying $t(x) = \mathsf{glue}$ on C_G and $t(x) = \mathsf{obstruction}$ on C_O, with no negative label inferred from certificate absence.

Corollary 2 (Direct-certificate identified sets)**.** *If $C_G \cap C_O = \varnothing$, the sharp identified set is*

$$I_C(x) = \begin{cases} \{\mathsf{glue}\}, & x \in C_G, \\ \{\mathsf{obstruction}\}, & x \in C_O, \\ \{\mathsf{glue}, \mathsf{obstruction}\}, & x \notin C_G \cup C_O. \end{cases} \tag{22}$$

Any conflict-free addition of direct certificates can only shrink these sets. If x remains uncertified, absence alone cannot point-identify obstruction. If $x \in C_G \cap C_O$, the direct authority record is contradictory and must be retained as CannotCheckConflict *rather than silently resolved.*

Proof. On a singly certified pair, every licensed completion has the certified value. On an uncertified pair, changing only $t(x)$ constructs licensed glue and obstruction completions with the same certificate observation, so the fibre criterion retains both values. Adding a nonconflicting certificate removes completions but adds none, hence identified sets shrink monotonically. □

This is a direct specialization of the general fibre and identified-set results, not a claim that direct ontology axioms exhaust semantic truth. Its role is to expose a finite authority domain on which binary scoring is licensed without converting nonselection into a negative label.

4.10 The sharp binary covering envelope

For a retained observation x, let $F_x = \{\omega : O(\omega) = x\}$ and let $I(x) = \{q(\omega) : \omega \in F_x\} \subseteq \{\mathsf{glue}, \mathsf{obstruction}\}$ be the binary identified set. An envelope rule E has distribution-free coverage on the licensed world class when $q(\omega) \in E(O(\omega))$ for every licensed world.

Corollary 3 (Minimal binary covering envelope)**.** *An envelope rule has distribution-free coverage if and only if $I(x) \subseteq E(x)$ for every attained observation x. Consequently I is the unique inclusion-minimal covering envelope. If one observation fibre contains both a glue world and an obstruction world, no singleton action is coverage-valid without an additional identification assumption.*

Proof. If E covers every licensed world and $y \in I(x)$, some $\omega \in F_x$ has $q(\omega) = y$; coverage gives $y \in E(O(\omega)) = E(x)$. The converse follows because $q(\omega) \in I(O(\omega)) \subseteq E(O(\omega))$ for every world. Inclusion-minimality is immediate. □

This is the standard identified-set argument specialized to the portrait decision, not a new general theorem. It makes the development boundary exact: agreement between correlated algorithms cannot by itself certify fibre constancy, while the full binary envelope attains coverage one by construction but may impose severe downstream deferral harm. A scientifically useful successor must license a proper subenvelope from additional, independently validated structure while retaining coverage.

4.11 Evaluation implied by the theory

The envelope theory changes the scientific target from forced-label accuracy to four jointly reported quantities:

1. *validity*: whether protected source-grounded truth is excluded from the returned envelope;
2. *sharpness*: the size or decision diameter of the envelope, reported only together with validity;
3. *floor-adjusted decision harm*: Equation (10) under a declared downstream loss; and
4. *information value*: the reduction in the identified set or decision floor from a specified observation refinement.

These targets envelop the original glue-or-obstruction calculus. A singleton envelope recovers an ordinary determinate mapping; a non-singleton envelope explains why unresolved is required; an empty joint-completion set certifies a global obstruction. The current experiments cover only special cases of this theory. They do not by themselves establish calibrated naturalistic envelopes or downstream utility.

4.12 Authority-augmented identification and composition

The portrait envelope above identifies content relative to an observation. Some scientific queries are additionally undefined unless the evidence possesses claim-relevant authority: a source and version are bound, the material may lawfully support the declared use, the method and comparator can execute, the population and semantics coincide, and the outcome is evaluated under the declared custody. Missing such a condition is not a third truth value and is not evidence against the method. It is an authority terminal.

For a query $q : \Omega \to \Theta$, freeze the finite set J_q of authority gates required by the claim and let $a_j : \Omega \to \{0,1\}$ be their truth in a world. Introduce $\perp_{\text{auth}} \notin \Theta$ and define

$$q^A(\omega) = \begin{cases} q(\omega), & \prod_{j \in J_q} a_j(\omega) = 1, \\ \perp_{\text{auth}}, & \text{otherwise,} \end{cases} \qquad \Theta_q^A(y) = \{q^A(\omega) : O(\omega) = y\}. \tag{23}$$

The terminal is the formal counterpart of CannotCheck. In particular, if all compatible completions lack a required executable or reference identity, then $\Theta_q^A(y) = \{\perp_{\text{auth}}\}$ rather than the empty set or a negative performance value.

Theorem 6 (Authority-aware fibre criterion)**.** *For a realised observation y and $\theta \in \Theta$, the authorized claim value is point identified as θ if and only if $\Theta_q^A(y) = \{\theta\}$. Equivalently, every authority gate is true and $q(\omega) = \theta$ throughout the observation fibre. If $\perp_{\text{auth}} \in \Theta_q^A(y)$, the observation does not identify the authority needed for the claim; if two ordinary values occur, it does not identify the claim's substance.*

Proof. $\Theta_q^A(y)$ is the image of the observation fibre under q^A. It is the ordinary singleton $\{\theta\}$ exactly when q^A is constantly θ on that fibre. By Equation (23), this holds exactly when all frozen gates equal one and q is constantly θ there. The two failure cases follow from whether the image contains the authority terminal or multiple ordinary values. □

The same refinement order applies. If O_2 refines O_1 while q and the frozen authority predicates are unchanged, the refined fibre is a subset of the coarse fibre and therefore

$$\Theta_q^{A,(2)}(O_2(\omega)) \subseteq \Theta_q^{A,(1)}(O_1(\omega)). \tag{24}$$

Truthful acquisition of a rights fact, executable revision or custodied outcome can therefore shrink the set. A mismatched repository, changed population or outcome-informed adapter is a changed scientific identity, not a refinement of the frozen observation.

Local readiness is not sufficient for global composition. Consider two worlds with the same observed stage artifacts and with every local source, rights, parser and entrypoint gate true. Let an unobserved relation say whether the two artifacts belong to the same scientific identity, with value zero in one world and one in the other, and let the end-to-end query be that relation. Every local statement is fixed and authorized, but the global identified set is $\{0,1\}$. A passed collection of local preflights therefore yields an end-to-end result only when the global query factors through those outputs under an explicit composition relation.

For comparator superiority, the required gates include both executable identities, resource and configuration completeness, use rights, task and population identity, reference authority, scorer custody and terminal- preserving completion. If any is absent or unresolved, a missing run cannot be imputed as a loss, tie, abstention, obstruction or zero. Native artifact smoke may identify a narrower executability query, but it does not identify superiority without the remaining truth, population and custody gates. This extends the content envelope to scientific evaluation without claiming a new general theory beyond the fibre criterion.

5 Public-reference mapping datasets

The empirical result in this paper uses two content-addressed *public-reference mapping datasets*. They contain already-structured scientific projections assembled from pinned external human and expert resources and from deterministic standards. They are not newly commissioned expert gold, and they do not evaluate raw-text extraction.

Confirmatory case family	**Cases**
Different name / same referent	13
Polarity, modality, attribution, or context	13
Valid / invalid representation mapping	6
Total	**32**

Table 1: Case-family composition of the disjoint confirmatory public-reference holdout.

5.1 Initial public-reference set

The initial mapping study contains 32 cases generated from pinned revisions of MUSE, SciFact, and SciSchema. The portable gold archive is bound by content digest, and the evidence package binds the external-source revisions and evaluator identity while retaining a checksum manifest rather than redistributing restricted third-party text. This first result is treated as development evidence; it is not pooled into the confirmatory verdict.

5.2 Disjoint confirmatory set

A second 32-case holdout was selected by an outcome-blind deterministic round-robin window over the sorted source pools, starting after the initial window and excluding all initial case identifiers. Its overlap with the first set is zero. Before any confirmatory system output was evaluated, the execution manifest froze the portable gold digest, the MUSE, SciFact and SciSchema revisions, evaluator component identities, bootstrap seed, margins, and terminal rule. Every digest named in that freeze is listed in Section 12.

Table 1 gives the composition. The 13 different-name/same-referent cases come from the MUSE cross-domain stratum. The 13 polarity/modality/attribution/context cases come from a scientific-claim-verification stratum. The six representation-mapping cases span materials, physics, and psychology and use deterministic standards where appropriate. Each record retains its source authority class and exact frozen input identity.

5.3 What the public-reference labels mean

The evaluated decision is a mapping-level terminal over already-structured projections. Depending on the source relation and the tested rule, the correct outcome can permit integration, forbid integration, or preserve an unresolved relation. The study evaluates false merges and false splits under these frozen cases. It does not claim that a model recovered the semantic coordinates from raw papers.

The confirmatory set is particularly discriminating for the family of polarity, modality, attribution and context cases: flat predicate canonicalization false-merges 6 of 13 cases (rate 0.4615), where coordinate-governed mapping makes no false merges. The different-name/same-referent and representation-mapping strata are clean controls on this holdout: all evaluated rules are correct there. That localization is reported rather than hidden, because the observed pooled effect is not evidence that every coordinate is necessary.

5.4 The broader expert-atlas design is not the dataset used here

A prospective expert-atlas design is preserved alongside this study: a protocol, annotation schema, handbook, adjudication policy, and 32 seed-to-gold templates spanning a broader eight-family design. Those templates are not expert gold. No independent dual annotation, per-coordinate inter-annotator agreement, specialist adjudication, or prospectively bound raw-text model run exists for that broader object, so its outcome remains undetermined and it is excluded from the claim made here rather than being presented retrospectively as the dataset used.

5.5 Data and licensing boundary

The public-reference package stores source identities, pinned revisions, locators, provenance and content commitments needed for replay. Third-party text is not vendored where redistribution is not permitted. A final submission archive will freeze the exact repository subject and stable public release URL for the legally shareable package; the absence of a permanent archive at the current drafting stage does not change the scientific result.

6 Evaluation

The empirical claim in this paper is governed by an execution-frozen protocol that was created after the initial 32-case result but before any confirmatory system output was examined. The predecessor result was allowed to motivate replication; it was not allowed to change the second holdout, the evaluator, the margins, or the pass criteria. The broader raw-text and expert-gold design remains a separate, unexecuted study. Both freezes are identified in Section 12.

6.1 Primary confirmatory hypothesis

The frozen replication hypothesis has two non-compensatory conditions:

False-merge superiority: the coordinate-governed-minus-flat false-merge point estimate must be at most -0.05, and the upper bound of the paired 95% bootstrap interval must be below zero.

False-split non-inferiority: the coordinate-governed-minus-exact-conservative false-split point estimate must be at most $+0.03$, and the upper bound of its paired 95% bootstrap interval must be at most $+0.03$.

The holdout passes only if both conditions hold. It contains 32 cases, has zero overlap with the initial 32-case set, and was bound by exact content digest before any confirmatory outcome was accessed.

6.2 Compared mapping rules

The confirmatory primary comparison uses three deterministic rules over the same already-structured mapping cases:

1. **Typed mapping calculus**: evaluates the scientific coordinates the case requires and can retain an obstruction rather than force a merge.
2. **Flat predicate canonicalization**: treats compatible surface predicates as sufficient merge evidence without the coordinate distinctions. This is the predeclared false-merge comparator.
3. **Exact-coordinate conservative control**: requires exact coordinate compatibility and can abstain where it lacks such a match. This is the predeclared false-split and non-inferiority control.

These are mapping-layer comparators, not attempts to reproduce a full external knowledge-graph, retrieval-augmented or schema-fusion system. Stronger schema-induction and provenance-contract baselines would be useful post-outcome construct-validity pressure, but they cannot rewrite a confirmatory status that was frozen beforehand.

6.3 Secondary ablations

The frozen mapping evaluator also reports descriptive paired ablations. The covered attacks are:

- **force compatibility without obstruction**: removes the explicit non-merge terminal and chooses a compatible mapping even where the coordinates prohibit it;
- **remove modality/polarity/attribution/discourse**: removes the coordinate group carrying the main confirmatory discrimination;
- **remove referent**, **construct**, **measurement**, and **temporal context**: retained as zero-effect controls on this particular holdout.

The ablations are secondary and descriptive. The protocol explicitly forbids turning their intervals into new confirmatory significance claims, and forbids reading zero effect on an under-supported stratum as evidence that a coordinate is unnecessary.

6.4 Metrics and statistics

The confirmatory decision uses case-level false-merge and false-split rates. Wilson 95% intervals describe individual rates. Paired differences use a 10,000-resample percentile bootstrap with frozen seed 20260817. The mapping rules are deterministic, so the protocol uses one execution per case rather than artificial stochastic repeats. Results are reported pooled and descriptively by case family and source stratum. The initial 32 cases are never pooled into the confirmatory verdict.

The primary decision rule is therefore fixed entirely by the two paired comparisons above. No post-outcome margin change, exclusion, reweighting, or replacement baseline can change whether the confirmatory run passed.

6.5 Custody and fail-closed rules

Before any confirmatory output was produced, the execution manifest bound the portable confirmatory gold digest, zero overlap with the predecessor gold, the exact MUSE, SciFact and SciSchema source revisions, the evaluator component identities, the bootstrap seed and pass margins, and a rule that any digest mismatch, source drift, evaluator drift or case overlap leaves the result undetermined rather than silently re-evaluating a different study.

The gold is not an input to the evaluated mapping rule. The evaluator consumes the candidate mapping result and the protected expected decision only after the system output has been sealed.

6.6 Reproducibility

The result is reproduced by deterministic public-reference scripts over immutable archives: the confirmatory analysis, publication figures and tables, source registry, evaluator identities, and an independent replay receipt are all retained, and two make targets regenerate the headline analysis and the publication artifacts from committed inputs. Section 12 names each of them.

This reproducibility claim does not extend to the unexecuted raw-text expert atlas, to full external retrieval or schema baselines, to generated-portrait recoverability, or to downstream answer quality. Each of those remains undetermined until its own prospective run exists.

6.7 Bounded method-structure pilot

The public-reference scientific-meaning experiment is not an evaluation of method-structure extraction, and is not offered as one. The larger expert-annotated method-structure corpus originally proposed for this extension—50–100 papers and algorithms across at least five fields, with independent annotation—was not available for this submission track and is therefore outside the empirical claim.

Instead, a prospectively frozen non-vacuity pilot uses six authored exact-ground-truth method pairs across numerical methods, graph algorithms and transactional workflows. The panel contains three structurally aligned pairs, two explicit obstructions and one source-insufficient case whose correct outcome is unresolved. Eight controlled corruptions delete or change an invariant, a reconstruction map, a dependency order, a precondition, failure semantics, lineage, an authority boundary, or an explicitly unknown progress coordinate. The typed representation and alignment path reproduces all six frozen pair labels and detects all eight material corruptions. A deliberately simple transcript and action-overlap comparator resolves only two of the six pair labels.

That supports a narrow claim only: the typed source-local objects are discriminating on a small exact-ground-truth panel and can preserve an explicit unresolved state. It does not estimate annotation agreement on real papers, false merge or split rates over natural scientific prose, end-to-end extraction quality, or the performance of any learner built downstream of these objects. Those remain undetermined until independently annotated evidence exists, and the pilot narrows the method-projection claim rather than establishing a general cross-domain method-learning result.

7 Post-saturation validation: representational compatibility is not scientific identity

The public-reference result answers a question about flat canonicalization. A stronger comparator makes a wider question testable: if a semantic integration stack has already produced well-structured candidate mappings, does that structure by itself authorize an assertion of scientific identity? After the public-reference result was frozen, a separate study was opened to test that without rewriting the original 32-case evidence. Its protocol was frozen before protected outcomes, and it credits scientific-variable decomposition, ontology and schema alignment, provenance-aware integration, plural and inconsistency representation, and generic consistency checking as prior work. The only disputed interface is whether those representational outputs authorize scientific identity on their own.

The scientific-identity contract battery. The protected battery contains 400 exact already-structured contracts: five scientific domains, eight mapping archetypes and ten disjoint variants per cell. The correct terminal for each contract is to glue, to record an obstruction, to record a plural view, or to leave the mapping unresolved.

Strong products. The primary comparator is granted strong structured semantic information: construct, measurement, context, provenance and explicit missingness. It correctly handles direct construct, measurement and context mismatches and missing load-bearing coordinates. Its systematic differences are three: it treats provenance as trace rather than as scientific stance, it forces a canonical choice for legitimate one-to-many mappings, and it accepts pairwise-compatible mappings without a global scientific-consistency check. A second comparator performs canonical matching and represents the usual merge-on-high-local-match pressure. A third is an ideal fully typed product that receives exactly the same coordinates and rules as the layer under test and is required to tie; it is an expressivity boundary rather than a weak baseline.

Protected result. The claim-relative compatibility layer reached exact scientific-integration decisions on 400/400 protected cases, against 250/400 for the strong semantic product and 50/400 for the canonical matcher. The paired difference against the strong product was +0.375 with a domain-stratified 95% bootstrap interval of $[0.3275, 0.4225]$; McNemar discordance was 150–0. The layer made no false merges, retained 1.0 clean-merge accuracy, and the paired effect was +0.375 in each of the five exact domains. An independent implementation reconstructed all four arms and the canonical protected-row digest exactly.

Boundary result. The ideal typed product also reached 400/400, with zero decision mismatches against the layer under test. The result therefore does not establish inherent expressivity or an advantage from centralization. What it supports is a bounded authority distinction: in the tested exact contracts, representational compatibility is weaker than scientific identity authority, and a claim-relative compatibility layer adds value over a strong semantic integration product by preserving obstruction, plurality and unresolved state where canonical identity is not licensed. The battery does not establish raw-text extraction, live ontology-engine superiority, expert agreement on new corpora, or open-ended cross-domain integration; each of those remains undetermined.

8 Successor interface: correspondence for representation candidates

The later ORION discovery programme can propose changes of representation, but Paper III treats such proposals first as mapping problems. A representation candidate must name the source and target objects, an explicit correspondence map, the obligations intended to be preserved, and the contracts the new representation is claimed to make available. These declarations are inputs to integration; they are not evidence that the correspondence is correct or that the representation should be scientifically adopted.

Paper III's local projection discipline supplies the relevant outcomes. A correspondence that preserves the registered source-local commitments can contribute GLUE; a mapping that drops a load-bearing assumption or invariant is an OBSTRUCTION; multiple defensible target readings may remain a PLURAL_VIEW; and missing or target-ambiguous reconstruction remains UNRESOLVED. A visually or terminologically similar representation is therefore insufficient when its assumption structure changes the scientific object being transported.

The merged zero-error regime-invention study supplies a useful negative boundary for this interface. On both the frozen primary and disjoint replication splits, the ORION-JUMP arm and the strongest verified representation-regime revision parent made identical structural decisions and achieved identical correspondence and protected-consequence outcomes, with zero false jumps. Its terminal is `REPRESENTATION_INVENTION_NO_INCREMENTAL_VALUE`. Thus the programme did not establish a distinct ORION representation-invention advantage beyond an already verified regime-revision parent. What remains load-bearing for Paper III is the explicit recoverability question:

when a representation changes, which scientific commitments are preserved, obstructed, plural, or reopened?

The same distinction controls transport of negative knowledge. A prior failure or exclusion can be reused across a surface-level representation change only when the correspondence establishes the required semantic coordinates and preserves the conditions on which that negative conclusion depended. If the target changes a load-bearing coordinate, the old exclusion becomes inapplicable or is explicitly reopened rather than copied as a global prohibition. If the mapping is incomplete, the transport claim stays unresolved.

Paper III therefore owns recoverable mapping, preservation and obstruction for successor representation candidates. It does not decide novelty, protected scientific validity or method adoption. In particular, a complete correspondence witness can make a later protected test interpretable, while the strongest-parent tie shows why such a witness must not be redescribed as a new general representation-invention mechanism.

9 Results

The evidence layers are deliberately kept separate. The public-reference projection-to-mapping protocol has been executed twice: an initial frozen 32-case study, followed by a disjoint, prospectively execution-frozen 32-case confirmation. The broader raw-text expert atlas, generated-portrait recoverability, downstream utility and full coordinate necessity are separate questions whose outcomes remain undetermined; they are not treated as missing pieces of the mapping result reported here. All numbers below come from immutable archived artifacts, listed in Section 12.

9.1 Initial public-reference mapping result

The public-reference builder used pinned external source authorities from MUSE, SciFact, and SciSchema and did not fill unsupported semantic coordinates with model guesses. The first deterministic build produced 32 already-structured mapping cases across five recorded strata and three case families: different-name/same-referent, polarity/modality/attribution/context, and valid/invalid representation mappings. A second freeze pass reproduced the portable gold artifact byte for byte.

On those 32 cases, the coordinate-governed comparison rule classified all expected relations correctly (accuracy 1.000), with false-merge rate 0.000 and false-split rate 0.000. Flat predicate canonicalization had accuracy 0.875 and false-merge rate 0.125, with no false splits. The paired false-merge difference was -0.125 with a 95% paired percentile-bootstrap interval of $[-0.250, -0.03125]$ (10,000 resamples; seed 20260817).

An exact-coordinate conservative control also produced zero false merges but abstained on 0.125 of cases and achieved accuracy 0.875. The initial result therefore motivated a prospective replication without itself becoming the final confirmatory authority.

9.2 Prospectively frozen confirmatory replication

A second, disjoint 32-case holdout was selected with the same outcome-blind round-robin construction at selection offset 32, excluding every initial case identifier. It was frozen before any confirmatory output of any compared rule was examined; the execution manifest bound its portable digest, upstream source revisions, evaluator identities, bootstrap seed, margins, and the terminal rule before evaluation.

The rule declared in advance required both (i) a coordinate-governed-minus-flat false-merge difference at most -0.05 with the upper end of its paired 95% bootstrap interval below zero, and (ii) a coordinate-governed-minus-exact-conservative false-split difference no larger than $+0.03$ with the interval upper bound no larger than $+0.03$. The holdout met both conditions. Coordinate-governed mapping had false-merge rate 0.000 and false-split rate 0.000. Flat predicate canonicalization had false-merge rate 0.1875 and accuracy 0.8125. The paired false-merge difference was -0.1875 with 95% interval $[-0.34375, -0.0625]$. The false-split difference against the exact-coordinate control was 0.000 with interval $[0.000, 0.000]$. The confirmatory primary verdict was therefore a **pass** under the rule frozen before those outputs.

The effect is localized to the strata that discriminate the tested rules. Among the 13 polarity/modality/attribution/context cases, flat canonicalization false-merged 6/13 (0.4615), where coordinate-governed mapping false-merged none. The 13 different-name/same-referent cases and the six representation-mapping cases were correct under all compared rules. These strata are reported descriptively and are not pooled with the initial 32 cases to inflate the confirmatory sample size.

9.3 Covered ablations

On the confirmatory holdout, forcing compatibility without the explicit obstruction distinction increased false merges by $+0.1875$ (95% paired bootstrap interval $[+0.0625, +0.34375]$) and reduced accuracy by the same absolute amount. Removing the modality/polarity/attribution/discourse coordinate group produced the same false-merge increase on the covered claim-verification cases.

Removing referent, construct, measurement, or temporal context has zero measured effect on this holdout. Those zero effects are preserved as coverage limitations, not read as proof that the coordinates are unnecessary or dispensable. The confirmatory protocol treats all ablations as descriptive secondary analyses rather than as new significance claims.

9.4 Separate-implementation verification and local falsifiers

A separately written implementation reproduced the confirmatory mapping result and its case-level decisions. This is implementation separation within the repository, not an external replication. The verification is bounded to the public-reference mapping claim; it does not promote the raw-text or downstream questions.

Synthetic exact-world tests separately exercise referent and construct identity, measurement equivalence, context, gluing, obstruction, plural views, preservation conditions, and typed mapping hypotheses. These remain engineering and semantic falsifiers rather than external empirical evidence.

9.5 Partial observation: a preserved adverse result and its bound

A staged partial-observation study asked whether adding an observedness-aware decision rule could repair over-resolution when one side of a mapping pair lost the coordinate that originally carried the decision. Mechanical redactions created 48 probes from the frozen atlases: 12 derivation cases, eight held-out public-reference cases, and 28 held-out exact cases. The current rule returned a determinate merge on all 48 although the redacted observation did not identify that decision. Rules that abstained on every asymmetric absence repaired this channel but destroyed determinate answers on intact cases. A more selective rule abstained only when admissible completions disagreed, improving on blanket abstention but still failing its zero-harm gate.

The last amendment showed why another observation-only rule cannot remove that cost. Its 36-case constructed source-record corpus contains 27 canonical observation orbits. Nine are two-case orbits in which the released projections are identical up to bookkeeping, identifier relabelling and left–right orientation, while the source-record gold relations differ. At least one member of each such orbit is therefore unrecoverable from the projections, so the maximum exact agreement of any candidate-visible deterministic rule is 27/36. The current rule reaches that ceiling with eight false merges and one false split. The selective unresolved rule avoids both error types by abstaining on both members of each ambiguous orbit, thereby losing the nine answers the current rule happens to get right there.

This is an exact identification result on a constructed corpus, not external evidence that natural scientific atlases have the same frequency or cost of missingness. The historical gates remain failed. The scientific consequence is the observation floor formalised in Theorem 3: a successor should compare avoidable false merge or downstream harm above that floor and should seek truthful observation refinements, rather than relabel a zero-harm threshold or claim that a more elaborate rule can recover erased information.

9.6 Public OAEI development: a preserved invalid-universe result

We next exercised one referent-alignment slice on the public OAEI 2004 bibliographic family, using AgreementMakerLight (AML) v3.2 as a source-native comparator. This is public development evidence, not the protected 768-cluster successor. The first identity failed before reference access: fragment-only matching created 2,153 ambiguous keys and one AML pair lay outside its explicit-declaration signature. We retained that failure. A distinct document-aware V2 identity then froze 68,187 type-compatible cases and 477,309 system predictions before joining the public references. AML executed 19 of 20 pairs with byte-identical output on replay; test 206 remained unparsable.

V2 failed both mandatory validity gates after the join. Its frozen candidate universe covered only 1,335 of 1,434 reference cells (recall 0.930962), and the conflict-preserving wrapper's gold-in-envelope coverage was 0.995542 rather than 1.0, with 304 envelope failures. Of these, 285 were missed public equivalences and 19 were asserted equivalences on public obstructions. The wrapper's exact rate was also 0.003710 below AML and its unresolved rate was 0.012187 higher. Its floor-adjusted harm was descriptively lower than AML in all three frozen loss regimes (differences -0.005430, -0.005723 and -0.039043), and an information-equivalent ideal tied it exactly; neither fact can promote an invalid evaluation universe.

The retained terminal is `PUBLIC_CANDIDATE_UNIVERSE_INVALID__PUBLIC_NONPROTECTED_ONE_SEED_FAMILY_ONLY`. The failure localises two successor problems: cross-construct public alignments require a broader input-only referent universe, and agreement between AML and label equality is not sufficient evidence for a point action. Missing panels, non-expressible ordered relations and shared evidence failure must remain CannotCheck. A new rule must attain envelope coverage 1.0 before its harm contrast is interpretable, and must then survive multiple untouched ontology families under protected custody.

9.7 Cross-construct V3 development: coverage repaired, harm not improved

After the V2 public gold had been inspected, we froze a distinct post-public-gold-informed V3 successor rather than rewriting the failed predecessor. Its input-only signature included explicit ontology entities plus named resources occurring in class- and property-valued positions, and it enumerated the full source-by-target cross product without construct-type blocking. The resulting universe contained 193,305 cases, of which 125,262 were cross-construct. The frozen V3 construction applied closed-world scoring only inside each public reference member's observed source–target domain; reference absence supplied its obstruction branch. Missing reference members, cases outside that domain, and non-equivalence relations were retained as CannotCheck, not imputed as obstruction. Because V3 design followed inspection of V2 public gold, it is development replay rather than confirmation.

The repaired universe contained all 1,399 public binary equivalence pairs, including 64 cross-construct equivalences omitted by V2, for candidate-universe recall 1.0. All 35 ordered non-binary relations were mapped to CannotCheck. Of the 193,305 cases, 117,914 were licensed for binary scoring and 75,391 remained CannotCheck: 2,415 lacked a reference member, 72,941 were outside the licensed reference domain, and 35 had non-binary relations. The full binary identification envelope attained gold-in-envelope coverage 1.0 and tied its information-equivalent ideal exactly. This yields the mechanics terminal `PUBLIC_V3_MAXIMAL_BINARY_ENVELOPE_COVERAGE_PASS`.

Coverage did not imply better action. The maximal envelope was unresolved on every scorable case and incurred mean harm 0.25 in each frozen regime. Candidate-minus-AML harm was +0.224799, +0.224732 and +0.193522; the comparative terminal was therefore `PUBLIC_V3_NO_HARM_SUPERIORITY`. The predecessor agreement-to-point rule also remained falsified on the wider licensed census: it produced 333 envelope failures, all cases in which AML nonselection and unequal normalized labels jointly licensed obstruction despite public equivalence. Thus V3 establishes a coverage–harm boundary, not candidate superiority.

The OAEI panel supplies descriptive input-identity contrasts rather than a causal proof for the referent coordinate. All 117,914 scorable pairs have different document-aware IRI keys, so there is no referent-no-change control; 75,512 also change construct type. AML v3.2 is reproducibly bound and its 19 successful outputs replay byte-identically, but test 206 remains unparsable, the execution used OpenJDK 17 rather than the upstream README's named Java 8 runtime, and no second current ontology matcher is frozen. The exact composite terminal is `PUBLIC_V3_MAXIMAL_BINARY_ENVELOPE_COVERAGE_PASS__PUBLIC_V3_NO_HARM_SUPERIORITY__PUBLIC_NONPROTECTED_ONE_SEED_FAMILY_ONLY`. Protected source-disjoint multi-family transport remains CannotCheck.

9.8 Outcome-blind V4 audit: binary truth is not yet identified

V4 was frozen as a distinct prospective source-family and comparator audit. It inspected metadata only and stopped before opening any new-family reference payload, relation value, matcher output or performance result. Seven OAEI track families had official track-page identities, but zero supplied the complete combination of explicit reference rights, provider-published cryptographic release identity and source authority needed to treat reference absence as binary obstruction. The predeclared minimum was three source-disjoint eligible families. Anatomy and LargeBio share NCI lineage, while Conference and MultiFarm share OntoFarm lineage; those pairs cannot be counted as four independent replications. The advertised 2025 Biodiv archive also returned HTTP 404 at the audit cutoff. The source panel therefore failed at 0/7.

This stop exposes a logically prior estimand failure. Under Theorem 5, a candidate omitted from a positive reference remains compatible with both glue and obstruction unless explicit negative evidence or authoritative exhaustivity removes one completion. The retained V3 join had 1,399 explicit glue labels and 116,515 obstruction labels produced by its residual absence branch. These counts do not change the V3 result or its exact composite terminal; they type its harm as conditional on the retained V3 closed-world construction rather than as source-certified binary truth.

The conditional selectivity frontier is also exact under that construction. At coverage one, a controller returning only correct singletons or the full binary envelope has mean harm $H = 0.25q$, where q is the full-envelope rate. Noninferiority to the strictest V3 AML regime therefore requires correct singleton selection on at least 89.919772% of cases. The sole frozen zero-failure positive cell selects 361/117,914 cases (0.306155%), yielding conditional projected harm 0.24923461. Sparse high-precision positive selection cannot resolve the V3 coverage–harm frontier.

Exact source identities are bound for AML v3.2, LogMap 4.0 and BERTMap through DeepOnto 0.9.3, but zero is execution-ready: family support, complete runtime and resource identities, native-artifact validators, positive-only adapters and the information-equivalent candidate/ideal manifest remain unfrozen. The joint V4 terminal is `PUBLIC_V4_SOURCE_RIGHTS_CANNOT_CHECK__PUBLIC_V4_SOURCE_FAMILY_DISJOINT_REPLICATION_CANNOT_CHECK__PUBLIC_V4_BINARY_OBSTRUCTION_SEMANTICS_CANNOT_CHECK__PUBLIC_V4_COMPARATOR_IDENTITY_CANNOT_CHECK`. Coverage, harm, noninferiority, superiority, protected transport and population generality were not tested.

9.9 V5 direct-certificate semantics calibration

A distinct outcome-blind V5 frame tested the finite authority rule rather than reopening the V4 OAEI source frame. Three immutable ontology families passed the separately audited rights, identity, direct-negative and conflict gates: ENVO at its frozen 2026-06-26 commit, FIBO Foundations at its frozen 2026Q2 commit, and the W3C PROV-O Recommendation. All 61 frozen Git blobs matched; 11,076,252 bytes parsed into 121,589 triples. No raw ontology payload was retained in the result packet.

Applying Corollary 2 yielded 4,838 conflict-free point-identified canonical pairs. Of these, 4,789 were GLUE: 4,780 same-IRI identities across byte-identical views and nine direct named `owl:equivalentClass` pairs. Forty-nine were OBSTRUCTION by direct named `owl:disjointWith` certificates. There were zero conflicts. Neither reference absence nor reasoner inference contributed a label. Every pair outside this finite certificate set retains the binary envelope unless separately certified.

The exact V5 terminal is `PUBLIC_V5_DIRECT_CERTIFICATE_SEMANTICS_CALIBRATION_PASS__NO_COMPARATOR_PERFORMANCE_EXECUTED__NO_NATURALISTIC_TRANSPORT`. AML, LogMap and BERTMap adapters remain 0/3 execution-ready, and no comparator, protected, harm or performance outcome was opened. Because each calibration pair uses byte-identical views of one frozen family, V5 validates the direct certificate semantics but does not test independently authored cross-ontology transport.

9.10 V6 comparator-native outcome-blind preflight

V6 tested only whether the three pinned comparator families could emit their required native artifact shapes on authored 16-class ontology fixtures. AML v3.2 exited zero in 0.319 seconds and emitted one parseable three-cell RDF alignment with source/target namespace guards. LogMap 4.0 exited zero in 1.015 seconds and emitted three equivalent RDF and TSV mappings. Its upstream RDF writer duplicated the ontology-1 IRI into both header positions; V6 retains this defect and requires row-namespace and RDF–TSV equivalence guards rather than treating the header as authority.

BERTMap through DeepOnto 0.9.3 did not reach training or prediction. The exact model weight hash matched, both 16-class ontologies loaded, and input-derived corpora reached 108 training and 12 validation records. Construction of `TrainingArguments` then failed because the pinned Python-3.10 lock selects Transformers 4.51.3, whose constructor exposes `eval_strategy`, while the pinned DeepOnto source passes `evaluation_strategy`. The process exited one before training and zero of five required mapping artifacts existed. This is CannotCheck, not an empty alignment or obstruction. A separately frozen V7 dependency-tuple successor has not been executed.

The exact V6 terminal is `P3_V6_TWO_OF_THREE_NATIVE_SMOKE_READY__BERTMAP_PINNED_LOCK_API_INCOMPATIBILITY_CANNOT_CHECK__V5_SCIENTIFIC_READINESS_UNCHANGED_ZERO_OF_THREE`. Two of three toolchains are native-smoke ready, but V5 scientific comparator readiness remains 0/3. No gold/reference alignment, protected case, scoring, harm, performance or naturalistic cross-ontology outcome was opened.

9.11 V7 BERTMap constructor and parser binding

A separately frozen, outcome-blind V7 successor binds the canonical BERTMap paper source at commit `ce848402b40e2f9513bf2d004894d3f82635022c` and maintained DeepOnto 0.9.3 at commit `74ca8d47f01bad0b8739f19ee2c392bdf6d9c090`. Under Python 3.10, a 26-distribution hash lock using Transformers 4.46.3, Tokenizers 0.20.3, Accelerate 1.0.1 and Torch 2.5.1 accepts DeepOnto's `evaluation_strategy` argument. This converts the V6 API mismatch only into `DEPENDENCY_API_CONSTRUCTOR_PASS`; no training, prediction or mapping is inferred.

The next native source stage exposes a different defect. When configured with a truthy threshold, DeepOnto's nonempty `EntityMapping` table reader iterates pandas namedtuples but indexes the row as `dp["Score"]`. A synthetic nonempty, non-reference row at that threshold therefore raises `TypeError: tuple indices must be integers or slices, not str`. An empty table does not enter the loop, and a falsey threshold short-circuits the defective comparison. Thus the V7 nonempty-reader diagnosis is valid only at the truthy-threshold boundary, not for every nonempty table. V7 also binds a closed five-artifact parser and passes 7/7 synthetic completeness, derivation, staleness and prohibited-field checks, but native BERTMap artifacts remain 0/5.

The exact V7 terminal is `P3_V7_BERTMAP_PAPER_SOURCE_AND_DEPENDENCY_CONSTRUCTOR_COMPATIBILITY_BOUND__CLOSED_FIVE_ARTIFACT_PARSER_BOUND__NONEMPTY_SOURCE_TABLE_READER_DEFECT_AND_FULL_NATIVE_SMOKE_CANNOT_CHECK__NATIVE_READINESS_TWO_OF_THREE__SCIENTIFIC_READINESS_ZERO_OF_THREE`. The source roots are Apache-2.0, but component-level rights and SBOM closure for the bundled Java and full Python/model/generated-artifact layers remain unbound. Native-smoke readiness stays 2/3 and scientific comparator readiness stays 0/3. No ontology, model, benchmark, gold, protected output, training, prediction, repair or scoring was opened or run.

9.12 V8 BERTMap table-reader source repair

V8 prospectively freezes a separately versioned minimal repair against the same DeepOnto commit and exact `mapping.py` SHA-256 `9cf0dce1c5bd142e4175f628f8f3267f54ed6deac9f31e165a25b4a073eedff0`. The patch changes only `dp["Score"]` to `dp.Score`; its SHA-256 is `412050c5d3a0e7891a21744a9690d3e4ba06886ea51eec273b6045bff688e42b`, and the resulting source SHA-256 is `d0b3b6cfdee45019783707c4bd623cc76f8325142828cf1e10ebb74ad628d70f`. After normalizing only that access, the method abstract syntax tree is unchanged.

Eight of eight outcome-blind synthetic cases pass. Empty tables and nonempty tables with a falsey threshold retain the pinned behavior. At a truthy threshold of 0.5, the pinned source reproduces the namedtuple `TypeError`, whereas the repaired source applies the existing threshold rule and retains only the 0.9 row. Missing or nonnumeric scores, stale source identity, reference-mode access and external-fixture access fail closed. The exact V7 parser accepts a temporary authored five-file shape with two raw, two extended, one filtered and one repaired row; those files were deleted and are not native BERTMap artifacts.

The exact V8 terminal is `P3_V8_BERTMAP_TABLE_READER_MINIMAL_REPAIR_SOURCE_HASH_AND_SYNTHETIC_EMPTY_NONEMPTY_EXECUTION_BOUND__MALFORMED_STALE_AND_PROHIBITED_CASES_FAIL_CLOSED__V7_PARSER_SYNTHETIC_COMPATIBILITY_BOUND__NATIVE_SMOKE_AND_SCIENTIFIC_READINESS_UNCHANGED`. The root source licence supports retaining the Apache-2.0 patch and notice, but the complete Python/JVM component rights and SBOM remain unbound.

Actual native artifacts remain 0/5, native-smoke readiness remains 2/3, and scientific comparator readiness remains 0/3. No DeepOnto import, JVM, model, ontology, benchmark, gold, protected output, training, prediction, mapping repair or scoring was opened or run.

9.13 V9 complete-runtime native attempt

V9 prospectively binds the complete Python/JVM runtime and passes its 30/30 component-rights gate before one offline native attempt. OpenJDK 17.0.19 starts; BERT fine-tuning, global mapping prediction, extension and filtering complete in 241.890 wall seconds. The repaired V8 truthy-threshold path is reached natively without the prior namedtuple `TypeError`. Four regular artifacts are retained: `raw_mappings.json`, `raw_mappings.tsv`, `extended_mappings.tsv` and `filtered_mappings.tsv`.

The final LogMap subprocess fails before producing the required repaired mapping file. LogMap 4.0 binds OWLAPI 4.1.3 and Guice 4.0; under Java 17, Guice cglib raises an inaccessible-object exception because `java.base/java.lang` is not open to the unnamed module. DeepOnto does not propagate that child failure and then raises a secondary missing-file error for the absent repair output. The unchanged five-artifact parser therefore returns `CANNOT_CHECK_NATIVE_ARTIFACT_CONTRACT_FAILURE`. No retry or post-result patch is made.

The exact V9 terminal is `P3_V9_COMPLETE_RUNTIME_RIGHTS_PASS__SINGLE_NATIVE_ATTEMPT_REACHED_FOUR_OF_FIVE_ARTIFACTS__LOGMAP_JAVA17_GUICE4_MODULE_ACCESS_CANNOT_CHECK__NO_RETRY__NATIVE_READINESS_TWO_OF_THREE__SCIENTIFIC_READINESS_ZERO_OF_THREE`. This closes runtime rights and replaces zero with four actual native artifacts, but does not close the required artifact contract. No gold, reference alignment, protected output, correctness, performance, harm, coverage or transport score is opened. The efficient next discriminator is a separately frozen repair-only microgate that keeps the exact Java/JAR/input hashes, adds only `--add-opens=java.base/java.lang=ALL-UNNAMED`, and requires direct child exit-zero propagation plus a regular repair output before any full rerun.

9.14 V10–V11 repair-only Java microgates

V10 executes the prescribed repair-only child once, but the retained thin LogMap JAR has zero of its 90 manifest-classpath dependencies after V9 cleanup. The child therefore exits one on a missing OWLAPI class before reaching the Guice module boundary. This leaves the efficacy of `--add-opens` CannotCheck; V10 is not retried and does not authorize a full run.

V11 is a materially distinct runtime-closure successor. It reconstructs all 90 adjacent JARs (25,337,399 bytes), verifies every hash against the V9 Java SBOM, and runs exactly one repair child with the same five launch identities and sole module-opening delta. The child exits zero in 0.429010 seconds and emits a regular 16-row repair file. The exact terminal is `P3_V11_EXACT_V9_CLASSPATH_90_OF_90_PASS__JAVA17_ADD_OPENS_REPAIR_MICROGATE_PASS__DIRECT_CHILD_EXIT_ZERO__REGULAR_REPAIRED_MAPPING_OUTPUT_PRESENT__NO_RETRY__FULL_NATIVE_SUCCESSOR_AUTHORIZED`. This is positive runtime and artifact-interface conformance only. The 16 source and target fields visibly retain `Optional.of(...)` wrappers, which V11 records without interpreting.

9.15 V12–V14 native closure, typed decoding and authority stop

V12 passes 15/15 pre-execution identity gates, including 126/126 distribution identities, 352/352 pinned source files, the V8 reader repair, V11's 90/90 Java closure, six model files and both ontology fixtures. Exactly one offline, no-gold BERTMap attempt finishes in 231.909885 seconds. The direct

LogMap child exits zero and all five required outputs are regular files with 16 rows. Native execution readiness therefore rises from 2/3 to 3/3.

The unchanged V7 parser nevertheless rejects the raw repair output because all 16 source and all 16 target strings are wrapped as `Optional.of(ontology-IRI)fragment` and hence are not literal members of the frozen role universes. V12 retains those bytes and labels structural conformance CannotCheck; it does not turn the artifact into an empty or incorrect mapping.

V13 freezes an anchored typed grammar over the retained rows. It performs no training, Java, LogMap, reference opening or scoring. Twelve of twelve adversarial malformed-wrapper cases pass; 16/16 source and 16/16 target values decode injectively to exact role-specific universe members; and the unchanged five-artifact parser exits zero. The decoded artifact SHA-256 is `c67ee88d541013f41984b239f9cdeaebdcd81573f2080d8af24c7688207dd0f3`, while the raw V12 repair remains byte-identical. This closes native shape and lexical-interface readiness, not mapping truth.

V14 then applies the cheapest reference-authority gate rather than scoring the authored fixture. It checks 21 admitted OAEI inventory units, 21 ontology-member hashes and 19 reference hashes. Neither frozen ontology hash nor either local `urn:orion` IRI matches provider evidence, and the registry contains no exact Bio-ML identity packet. V14 therefore opens no gold and computes no precision, recall, F1 or comparator score. Its exact terminal is `P3_V14_PROVIDER_NATIVE_REFERENCE_IDENTITY_CANNOT_CHECK__FROZEN_PAIR_IS_SYNTHETIC_AND_MATCHES_NO_OAEI_BIOML_CASE__NO_PUBLIC_GOLD_ADMITTED__NO_METRICS_OR_COMPARATOR_COMPUTED`. Scientific comparator readiness remains 0/3 pending a separately frozen, provider-native identity packet and prospective evaluation.

9.16 V15–V21 provider-native execution and one-case comparison

V15 closes the V14 identity gap only for a separately named OAEI 2004 test-103 successor. It binds the Zenodo record and DOI, the exact source, target and public-reference hashes, and AML v3.2 under their provider licences; the adverse V14 statement about the synthetic fixture remains true. V16 then runs AML once under the upstream-named Java 8 environment, without supplying a reference. The process exits zero in 2.295929 seconds and freezes one well-formed 46-cell alignment before semantic reference access.

The first BERTMap attempt on the same provider-native pair does not produce a mapping. V17 passes 30/30 outcome-blind preflight checks, then HermiT rejects a transitive `isPartOf` property used inside a maximum-cardinality restriction. The one attempt ends after 89.887939583 seconds with zero of five required artifacts, no retry and no reference access. Its exact terminal is `P3_V17_BERTMAP_NATIVE_ATTEMPT_FAIL__NO_RETRY__COMMON_SCORING_NOT_AUTHORIZED`. V18 tests a structural-reasoner route. Both ontologies load, both consistency queries return true and the required taxonomy operations succeed, but the gate incorrectly expects 36 class keys and 33 annotation keys where DeepOnto exposes 37 keys for each surface. It therefore retains `P3_V18_STRUCTURAL_REASONER_COMPATIBILITY_FAIL__V19_BERTMAP_NOT_AUTHORIZED`; this is a failed surface contract, not evidence that the structural reasoner is incompatible.

V19 isolates that surface assumption before any matching or gold access. Each raw runtime dictionary is exactly the frozen 36-class role universe plus `owl:Thing`. Removing only that built-in from the Python dictionary, without changing ontology axioms, yields the exact 36-by-36 matcher universe; each annotation index retains 36 keys, of which 33 have nonempty local labels and three external keys are empty. V20 then runs the V19-authorized structural entrypoint once. It exits zero after 281.646897292 seconds, its direct LogMap child exits zero, all five native artifacts are regular files and the repaired table contains 33 rows. The inherited V13 grammar nevertheless rejects those rows because V20 emits plain absolute IRIs rather than the earlier `Optional.of(...)` surface. V20 preserves the exact error `ValueError:_source_surface_string_violates_frozen_typed_grammar` and the terminal `P3_V20_BERTMAP_NATIVE_PASS__TYPED_DECODER_OR_STRUCTURAL_CONTRACT_FAIL__COMMON_SCORING_NOT_AUTHORIZED`; the reference remains unopened.

V21 applies a distinct post-output, pre-reference interface to the immutable V20 artifacts. It admits only whole-string absolute IRIs that are exact, injective members of the frozen source and target role universes; decoding is the identity transform, with no trimming, substitution, row deletion or row addition. All 33 rows pass, and the unchanged structural parser passes with 36/36 expected source keys. Only then does the frozen evaluator open the public reference and score both already frozen outputs. The analysis unit is one OAEI 2004 test-103 case, not its pair cells; consequently we report exact finite-case descriptions and no confidence interval, p value or population estimand.

Table 2: Exact descriptive pair metrics on the single frozen OAEI 2004 test-103 case. The primary common-class estimand compares the opportunity shared by both systems; the secondary full-task estimand keeps BERTMap's absent property-mapping coverage visible. The case is the analysis unit ($n = 1$); pair cells are not independent replicates, and no confidence interval or p value is computed.

Estimand	System	Pred.	TP	FP	FN	Precision / Recall / F1
Common class pairs	AML v3.2	8	8	0	25	1, 8/33, 16/41
Common class pairs	BERTMap	33	33	0	0	1, 1, 1
Full equivalence pairs	AML v3.2	46	8	38	83	4/23, 8/91, 16/137
Full equivalence pairs	BERTMap	33	33	0	58	1, 33/91, 33/62

The exact BERTMap-minus-AML F1 differences are 25/41 on the primary common-class estimand and 3529/8494 on the full equivalence-pair estimand. Both favour BERTMap on this case. They do not authorize a general-superiority, current-state-of-the-art, protected-confirmation or naturalistic-transport claim, and the 58 full-task false negatives prohibit presenting the class-only score as complete ontology-alignment performance. V22 therefore requires at least seven source-disjoint cases spanning at least two provider or track families and two ontology domains, with fixed seeds and thresholds, independent custody, both estimands, a current stronger comparator where interfaces permit, and worst-case as well as aggregate gates.

9.17 V7T authority-composition result

The V6 split between 2/3 local smoke readiness and 0/3 scientific readiness is not only a reporting convention. Section 4.12 lifts each claim into an authority-augmented identified set: missing source, rights, execution, reference, population or custody authority produces a distinct terminal outside the scientific truth codomain. The authorized value is a singleton exactly when every frozen authority gate is true and the substantive query is constant throughout the observation fibre.

A two-world finite countermodel makes the composition gap explicit. Both worlds have the same observed stage artifacts and pass the same local source, rights, parser and entrypoint gates, but they differ in an unobserved relation stating whether the artifacts share one scientific identity. The end-to-end query therefore retains the identified set $\{0, 1\}$. Passed local preflights compose only through an explicit cross-stage relation; they cannot be counted as a comparator loss, tie or score. The exact terminal is `P3_V7T_AUTHORITY_AUGMENTED_FIBRE_CRITERION_PROVED__LOCAL_READINESS_NONCOMPOSITION_COUNTERMODEL_PASS__V6_SCIENTIFIC_READINESS_U`

NCHANGED_ZERO_OF_THREE__NOVELTY_CANNOT_CHECK. The result is mathematical. It opens no matcher output, protected truth or performance result, and distinct theorem priority remains CannotCheck pending an exhaustive independent review.

9.18 What this result does not establish

The reported result does not establish:

- **Raw-text end-to-end superiority:** no prospectively bound raw-text extractor or model comparison against strong retrieval-augmented or schema systems supports this paper's headline;
- **Generated-portrait recoverability or downstream utility:** no frozen reader-level or downstream task result is promoted here;
- **Naturalistic envelope calibration:** the constructed partial-observation bound proves an observation-level impossibility on its frozen cases, but does not estimate the prevalence, sharpness or loss of partial identification in external scientific corpora;
- **A multi-case OAEI comparative result:** the public one-family V2 identity failed candidate-universe recall and envelope-coverage gates; V3 repairs binary coverage only by returning the maximal unresolved envelope and is worse than AML in every frozen harm regime; V4 opens no new-family outcomes because none of seven candidate families passes its rights, immutable-identity and binary-truth gate; V5 supplies a finite direct-certificate calibration but opens no matcher outcome; V21 adds exact same-universe scoring on one historical test-103 case, but no source-disjoint multi-case or population comparison;
- **Universal coordinate necessity:** targeted cases for referent, construct, measurement, temporal context and additional obstruction families do not exist in these datasets;
- **An executed expert eight-family gold:** the broader seed templates and annotation protocol remain prospective rather than being relabelled as the public-reference dataset.

Each of those outcomes therefore remains undetermined, and stays visible as undetermined so that the replicated mapping result cannot be read as authority for the wider design. Any future positive has to arrive as a new prospectively frozen artifact rather than as a widening of the current claim in prose.

10 Limitations and ethics

10.1 Limitations

Theory–evidence separation. The fibre, information-refinement, positive-reference non-identification and robust-decision results are mathematical statements under their declared assumptions. So is the pairwise-consistency counterexample. Their validity does not depend on the sample sizes below; their scientific usefulness does. The framework does not yet show that a deployed extractor can construct a valid and sharp naturalistic portrait envelope, that its admissible-world model contains the relevant source relation, or that experts will agree on the constraints defining that model. A misspecified envelope can be confidently narrow and wrong. Empirical coverage, sharpness and assumption sensitivity must therefore accompany any future application.

Decision losses are not scientific facts. The merge/separate/defer bound makes costs explicit but does not estimate them. False-merge, false-split and deferral costs depend on the downstream task and stakeholders. A mapping mechanism should expose identified sets and provenance; it should not launder one author's utility function into a universal scientific relation. No downstream cost elicitation or protected decision study has yet been executed.

What is and is not owned here. Almost every mechanism this work depends on is prior art, and the boundary is worth stating once rather than repeatedly. Structured scientific-variable semantics, ontology and schema alignment, provenance modelling, provenance-aware pluralism, evidence-linked schema induction and fusion, entity resolution, stance and contradiction analysis, retrieval-augmented synthesis and generic consistency checking are all donor mechanisms, described and cited in Section 2. The residual is the claim-relative decision that runs after all of them: which coordinate differences license an assertion of scientific identity, and which oblige the record to keep an obstruction, a plural view or an unresolved relation visible instead. A result in favour of that decision layer is not a result in favour of the donor mechanisms, and none of the evidence here should be read as ownership of them.

Scope of the public-reference result. The reported result is the public-reference projection-to-mapping study, not an end-to-end expert atlas or an independently curated naturalistic panel. It contains two separately frozen 32-case mapping sets assembled from pinned public human and expert resources; the confirmatory set is disjoint at the case level from the initial set. These externally sourced relations were transformed into host-constructed, already-structured scientific projections. They therefore test mapping and obstruction semantics, not whether a model can reliably extract all required semantic coordinates from raw papers.

Scope of the post-saturation battery. The scientific-identity contract battery is a separate, prospectively frozen exact-contract study, not a pooled extension of the 32-case public-reference result. Its five domain labels denote heterogeneous scientific mapping contracts with mechanically defined ground truth; they are not executions of five live ontology or schema engines, and they do not substitute for newly commissioned expert agreement. The battery supports the bounded decision principle that representational compatibility can be weaker than scientific identity authority in the tested contracts. It does not establish ontology-engine-agnostic superiority or open-ended cross-domain integration.

Expert subjectivity and theory-ladenness. Scientific identity and measurement-equivalence judgments are theory-laden and may depend on disciplinary background. The prospective expert-atlas design mitigates this through a written handbook, independent double annotation, coordinate-wise agreement reporting, domain-expert escalation, and an explicit unresolved state. That expert-gold study has not been executed, so this manuscript reports no inter-annotator agreement and presents no seed template as expert gold.

Domain and case-family coverage. The public-reference route covers only the case families its pinned source authorities support. Its confirmatory effect is concentrated in polarity/modality/attribution/context cases; different-name/same-referent and representation-mapping cases are useful controls but do not discriminate the compared rules on that holdout. The successor battery broadens the exact contract families but still does not establish universal cross-domain adequacy or the necessity of every semantic coordinate. In particular, a coordinate's usefulness in the exact successor does not imply that every scientific domain must encode that coordinate in the same representation.

Evolving terminology. Scientific terminology and operationalization change over time. The mapping study is bound to specific source revisions and does not establish robustness to temporal drift, historical conceptual change, or cross-lingual terminology.

Ontology and schema dependence. The semantic-coordinate representation is a design choice. Other representations may encode additional causal, probabilistic, temporal, or normative structure. The current evidence supports a bounded typed mapping and obstruction result and a bounded claim-relative compatibility successor; it does not establish that the chosen coordinate system is uniquely correct or optimal. An ideal product given the same scientific coordinates and rules ties the successor layer exactly.

Extraction error propagation. The text-to-scientific-meaning extraction layer remains unexecuted. Extraction errors could propagate into mapping, gluing and obstruction decisions. Because no prospectively bound raw-text extractor run supports the headline result, this paper reports no stage-attributed extraction error and claims no end-to-end superiority over raw-text, retrieval-augmented or schema systems.

Recoverability and downstream utility remain open. The design intends global statements to retain source projections, mapping paths, preservation conditions, and non-preserved coordinates. Local tests verify those mechanics, but the current external and exact-contract studies establish neither reader-level recoverability of generated portraits nor downstream scientific-answer improvement. Both remain undetermined pending their own frozen evaluations.

The external partial-identification successor is unexecuted. A distinct prospective successor specifies a target of 768 independent source-artifact-family clusters across 16 strata and four domains, requires nonzero within-comparison contrast in referent, construct, measurement and temporal-context coordinates, and defines its endpoint as avoidable false merge or downstream decision harm above the proven observation floor. The first outcome-blind metadata preflight fixed twelve GitHub software roots. Its provider-diverse successor now fixes 16 candidate roots across two waves, with four metadata providers, four artifact modalities, four nominal domains and two roots per domain in each wave. Thirteen of thirteen conformance checks and all 16 live identity checks pass, but no case text was opened and no coordinate, pair or gold label was assigned. Eight roots lack exact historical Crossref bytes; content-class rights and case eligibility remain unresolved for all sixteen. This is a structurally diverse candidate frame, not the 768-cluster atlas. The successor's four-comparator field is also a count rather than four named systems with revisions, licences, adapters and terminal maps. The original frozen binding says "nonzero variation", but Theorem 4 shows that between-block variation can coexist with zero comparison opportunity. The outcome-blind amendment therefore interprets claim-bearing opportunity as within-comparison contrast and additionally requires joint-coordinate support to be reported; it changes no endpoint, sample size, outcome, or historical terminal. The required source-disjoint external atlas, independent protected gold, raw-text attack corpus, downstream-evaluator custody and comparator family therefore do not yet exist. The current terminals are `P3_PARTIAL_IDENTIFICATION_READY_FOR_EXTERNAL_ATLAS_NOT_A_SCIENTIFIC_RESULT` and `P3_COMPARATOR_FAMILY_CANNOT_CHECK_IDENTITY`; no outcome is reported.

The public OAEI development identity is adverse, not protected evidence. The OAEI 2004 run covers one systematic bibliographic seed family with public references. The immutable V2 predecessor omitted 99 of 1,434 reference cells under its frozen definition, and its wrapper excluded the public gold in 304 of 68,187 cases. Those remain validity failures even though its descriptive floor-adjusted harm contrast against AML was negative in three loss regimes. A distinct gold-informed V3 successor removes the construct block, includes all 1,399 binary equivalence pairs, maps 35 ordered relations to CannotCheck, and attains envelope coverage 1.0. That mechanics pass is vacuous as a performance result: the maximal binary envelope is unresolved on all 117,914 scorable cases and is worse than AML in each frozen harm regime by +0.224799, +0.224732 and +0.193522. The agreement-to-point rule still fails 333 times, all on public equivalences licensed as obstructions by shared negative evidence.

Neither development identity supplies an independent family-level sampling distribution, protected labels, causal coordinate separation, temporal or plural-truth opportunity, a current strongest-comparator set, or evaluator custody. AML was replayable on 19 pairs, but test 206 remained unparsable and the Java runtime differed from the upstream named version. The OAEI evidence therefore cannot authorize a confidence interval, cross-domain comparison, protected confirmation, PLURAL or temporal claim, or any conclusion for the frozen 768-cluster target. The next discriminator is a proper selective subenvelope on multiple untouched ontology families with binary coverage 1.0, harm noninferiority in every frozen regime, current strong comparators and cluster-level replication.

V4 is a pre-outcome feasibility stop, not a performance result. The seven-family audit was metadata-only. Finding no complete record of rights, immutable release identity and binary-negative authority does not prove that every audited reference is incomplete; it means the prospective V4 protocol could not license closure or open outcomes. No new-family coverage, harm, noninferiority, superiority or generalization quantity was observed. Likewise, binding exact source identities for AML, LogMap and BERTMap does not make them execution-ready or establish which comparator is strongest. The 89.919772% selection requirement and projected harm 0.24923461 are arithmetic conditional on the V3 loss and V3 binary construction, not transported family performance. Protected evidence, evaluator custody, coordinate value and population transport remain CannotCheck.

V5 calibrates direct authority, not naturalistic transport. The three admitted families use byte-identical source and target views, so same-IRI identity is intentionally abundant and no independently authored alignment shift is present. The 4,838 direct certificates identify only their finite canonical pairs; they do not make the remaining Cartesian complement negative. AML, LogMap and BERTMap remain 0/3 execution-ready, no matcher or protected outcome was opened, and no V3 harm result is reversed. The next discriminator is an outcome-blind native-artifact preflight followed by at least three independently governed, independently authored ontology-pair families that pass the unchanged rights, identity, explicit-negative and conflict gates.

V6 closes two native smoke surfaces, not comparator readiness. AML and LogMap emitted parseable native artifacts only on authored fixtures; the rows were not compared with gold and say nothing about correctness, coverage, harm or transport. LogMap's duplicated ontology header is an upstream metadata defect guarded by row namespaces, not silently repaired. BERTMap produced no required mapping artifact because the frozen DeepOnto and Transformers APIs are incompatible before training. Missing BERTMap artifacts remain CannotCheck, never a negative alignment. The separately frozen V7 successor repairs the constructor mismatch and binds a closed five-artifact parser. Its downstream failure is narrower than first reported: only a nonempty read with a truthy threshold reaches string indexing on the pandas namedtuple; falsey thresholds short-circuit. V8 repairs that one expression and passes 8/8 synthetic language-level cases. V9 then binds complete Python/JVM rights and reaches the repaired branch natively, but emits only 4/5 required artifacts: Java 17 blocks Guice 4.0 reflective access during LogMap repair, so the repaired mapping is absent. V10 then shows that the retained thin JAR alone cannot test the module-opening repair because its 90-entry manifest classpath is missing. V11 reconstructs that exact closure and passes the repair microgate; V12 completes one full no-gold run with five regular artifacts; and V13 maps the preserved `Optional.of(...)` strings injectively into the exact role universes under a frozen typed grammar. These steps raise native execution readiness to 3/3 and close the structural parser, but none is a comparator result. V14 shows why: the authored pair is synthetic and matches no admitted provider-native reference identity. Scientific readiness remains 0/3 until a provider-native pair, exact reference and same-universe comparators are frozen prospectively under protected custody.

V21 is a one-case positive, not a population comparison. V15–V21 close the provider-native identity, exact-Java-8 AML execution, structural BERTMap execution and same-universe scoring chain for one OAEI 2004 test-103 case. The intervening failures remain part of the evidence: V17's HermiT attempt emitted no artifacts, V18's surface contract miscounted `owl:Thing` and empty annotation keys, and V20's inherited wrapper grammar rejected plain direct IRIs. V19 and V21 repair only those causally localized interfaces; neither rewrites an ontology axiom or matcher row.

The resulting class-pair F1 difference of 25/41 and full-task F1 difference of 3529/8494 favour BERTMap on the enumerated case. They do not estimate a family- or population-level effect. Test 103 belongs to one historical bibliographic seed family, same-workspace execution is not independent custody, and neither thresholds nor model revision have been tested across source-disjoint cases. Moreover, the primary estimand excludes property pairs because BERTMap exposes no property-matching opportunity; the mandatory full task leaves 58 reference pairs unmatched. Pair cells are not independent replicates, so no p value or confidence interval is reported. A publishable generality claim requires the V22 multi-case, multi-family, multi-domain and independent-custody design rather than repetition or inferential inflation of this finite result.

10.2 Ethics

The public-reference study reuses published research resources and collects no new human-participant data. The successor battery uses exact authored contracts rather than new human annotations. The broader expert-annotation protocol is prospective and has not been executed. An obstruction or a plural view should be presented as a provenance-bound record of incompatibility or of unresolved mapping, not as a judgment that one scientific perspective is intrinsically superior. Source lineage is retained so that a downstream reader can inspect the evidence and the conditions behind an integration decision.

11 Conclusion

A global scientific portrait should not be forced to say more than its sources identify. This paper replaces the single canonical portrait with an epistemic portrait envelope: all global portraits compatible with the observed source-local projections, provenance and licensed mapping constraints.

The change is not a retreat to vagueness. It produces exact conditions for when a claim may be sharp and exact bounds for acting when it cannot be.

The fibre criterion proves that a query is point identified precisely when it is constant across every admissible world yielding the same observation. If two worlds share the observation and require different answers, no rule over that observation can be uniformly correct. The universal factorisation theorem identifies the implementation-neutral target: every sufficient interface must refine the query's identified-set map, and information-equivalent products are required to tie. The information-refinement theorem then identifies the constructive route upward: truthful source acquisition, extraction, measurement or provenance can shrink the envelope and weakly lower the minimax decision floor. More elaborate inference over unchanged evidence cannot. The coordinate-support theorem adds the corresponding design boundary: variation between unrelated cases does not identify coordinate value when the coordinate never contrasts inside a registered comparison, and nonzero contrast still requires joint-support analysis when coordinates co-move. A binary robust-decision result states when merge, separation or deferral minimises worst-case loss, and the joint-completion result shows why pairwise-compatible mappings still require a global consistency check. The positive-reference non-identification theorem adds the evaluation boundary: omission cannot be converted into obstruction unless explicit negative evidence or authoritative exhaustivity over the frozen candidate universe removes the otherwise compatible glue completion.

These results unify four terminals. Glue is licensed when the relevant claim is invariant across the feasible envelope. A typed obstruction is an identified failure of joint compatibility. A plural portrait represents multiple source-valid views inside the scientific object. Unresolved state records variation across feasible completions. In particular, scientific plurality and analyst ignorance are not the same state.

The evidence supports narrower instances. On a disjoint 32-case confirmatory holdout assembled from pinned public human and expert resources, the typed comparison made no false merges, while flat predicate canonicalization false-merged 0.1875 of cases; the paired difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. Against the exact-coordinate conservative control, the false-split difference was 0.000 with interval $[0.000, 0.000]$. A separate 400-contract exact battery produced 400/400 exact decisions for the claim-relative layer against 250/400 for a strong semantic product; an ideal product receiving the same coordinates and rules tied at 400/400. The tie rules out inherent expressivity or centralisation as the explanation.

The most informative result is adverse. In the constructed partial-observation corpus, nine two-case orbits share exactly the evidence visible to every compared rule while carrying different source-record gold. No observation-only rule can exceed 27/36 exact answers. The current rule attains that ceiling by accepting eight false merges and one false split; the selective unresolved repair avoids those determinate errors only by giving up the nine answers the current rule gets right in the same orbits. The failed historical gate is preserved. Its successor asks the scientifically stronger question: which system reduces *avoidable* harm above that observation floor after independent, source-disjoint observation and gold are supplied?

A public OAEI development run makes that question more concrete without answering it. The conflict-preserving wrapper showed lower descriptive harm than AML in three frozen regimes, but the candidate universe covered only 1,335/1,434 reference cells and the envelope missed the gold on 304/68,187 cases. The invalid-universe terminal is therefore retained. A distinct gold-informed V3 successor represents cross-construct pairs, contains all 1,399 public binary equivalences, maps 35 ordered relations to CannotCheck, and reaches envelope coverage 1.0. It does so only by returning the full binary envelope on every scorable case. Its harm exceeds AML by $+0.224799$, $+0.224732$ and $+0.193522$, while the predecessor point rule still fails 333 times on public equivalences. Thus complete coverage is attainable on this public family, but no nontrivial selective envelope with acceptable action cost has been shown. The next successor must add independently validated structure, retain coverage and harm noninferiority, and transfer to multiple untouched ontology families under protected custody.

The outcome-blind V4 audit did not yet supply that transport. Zero of seven candidate OAEI track families passed the complete rights, immutable-identity and binary-truth gate, against three required, so no new-family outcome was opened. The V3 join's 1,399 explicit glue labels and 116,515 absence-derived obstruction labels therefore support V3 only under its retained closed-world construction. Under that same construction, coverage-one noninferiority to the strictest AML regime requires 89.919772% correct singleton selection, whereas the sole frozen zero-failure positive cell selects 0.306155%. Exact AML, LogMap and BERTMap source identities are now bound, but all three remain execution-unready. The V4 contribution is thus an authority-aware non-identification result and a sharper next discriminator, not a positive performance or superiority claim.

V5 closes one narrower semantic bridge without rewriting V4. Three rights-audited immutable families passed the direct-certificate gate, yielding 4,838 point-identified pairs: 4,789 glue and 49 explicit obstruction, with zero conflicts and no absence-derived label. This supports the general rule that licensed direct certificates refine the binary envelope monotonically while uncertified pairs remain set-valued. Because the calibration uses byte-identical views and all three comparator adapters remain execution-unready, it is neither naturalistic cross-ontology transport nor a comparator result.

V6 advances the interface boundary without changing that scientific terminal. AML and LogMap now pass guarded native smoke on authored fixtures; BERTMap stops before training because the pinned DeepOnto/Transformers APIs disagree, leaving zero of five required mapping artifacts. The resulting two-of-three smoke count is not two comparator results: no output was compared with gold, and no protected or naturalistic pair was scored. Scientific comparator readiness remains 0/3. The separately frozen V7 successor repairs the BERTMap constructor mismatch and binds a closed five-artifact parser. V8 then repairs the exact truthy-threshold table-reader access with a one-expression source patch and passes 8/8 synthetic language-level cases. The predecessor's broader wording is corrected: falsey thresholds short-circuit the defective subscript. V9 binds the complete runtime and reaches that repair in one native attempt, producing 4/5 artifacts after fine-tuning, prediction, extension and filtering. The missing repair artifact is localized to Java 17 module access in LogMap 4.0's Guice 4.0 path; no retry is made. V10 localizes an independently missing 90-entry manifest classpath, and V11 reconstructs that exact closure so the repair-only child exits zero. V12 then completes one full no-gold run with five regular artifacts, raising native readiness to 3/3. V13 preserves the raw wrapped repair, decodes all 16 source and target identifiers injectively under a frozen typed grammar, and passes the unchanged structural parser. V14 refuses to score because the synthetic fixture matches no admitted provider-native reference identity. Scientific readiness remains 0/3: interface closure is not mapping truth, correctness, performance, harm, coverage or transport.

A separately identified provider-native successor closes one finite comparator case without changing those historical terminals. V15 binds OAEI 2004 test 103 and its public reference; V16 freezes an exact-Java-8 AML output; V17 preserves HermiT's no-artifact failure; V18–V19 localize and repair only DeepOnto's built-in class surface; and V20 freezes a successful 33-row native BERTMap output while preserving the inherited decoder failure. V21's exact identity-transform decoder then authorizes same-universe scoring. BERTMap's F1 is 1 versus AML's 16/41 on the common 33-class opportunity and 33/62 versus 16/137 on the full 91-pair task. The second estimand also exposes 58 unmatched property pairs. Thus this chain establishes a genuine positive on one enumerated historical case, not general superiority. The next scientific step is the prospectively frozen, source-disjoint V22 panel with case-level analysis, independent custody and worst-case gates.

The authority-augmented fibre result explains why that split is stable under claim widening. A local parser or smoke pass identifies only its local query. An end-to-end scientific comparison additionally requires constant substantive truth and source, rights, population, executable, reference and custody authority over the whole observation fibre. A two-world countermodel shows that all local gates may pass while one unobserved cross-stage identity relation leaves the global query at $\{0, 1\}$. Thus local readiness cannot be added into scientific authority without an explicit composition relation, and a missing comparison cannot be imputed as a loss, tie or zero.

The resulting programme is wider than a coordinate mapper but does not borrow authority from unexecuted studies. Partial identification, robust decision theory, scientific-variable semantics, ontology and schema alignment, provenance, constraint processing and pluralism are donor theories. The proposed synthesis turns their outputs into a claim-relative envelope and a downstream decision interface. Naturalistic envelope validity, raw-text robustness, independent multi-domain gold, source recoverability and measured downstream gains remain open. The sixteen-root provider-diverse successor is a metadata-only candidate frame with unresolved content-class rights and zero assessed eligible cases, and the four-comparator field is still a count rather than an identity freeze; neither constitutes an atlas, a comparator result or external custody. Those are now explicit research problems with identifiable estimands and falsifiers, not reasons to narrow the theory or to rename missing evidence as success.

12 Data availability

The repository snapshot contains the frozen public-reference datasets, successor evidence and content bindings used by this study. Every artifact named anywhere in this paper is listed here with its path, so that no repository location has to be carried in the narrative. Paths below are relative to the paper directory unless they begin with `src/` or `research/`.

12.1 Preregistration

The end-to-end study was preregistered and frozen before any outcome was accessed, under the identifier `P3.cross-domain-atlas.v1`. Its machine-readable freeze is `protocol/PROTOCOL_V1.json`, with the annotation schema, handbook and adjudication standard in `protocol/ANNOTATION_SCHEMA_V1.json`, `protocol/ANNOTATION_HANDBOOK_V1.md` and `protocol/ADJUDICATION_POLICY_V1.md`. That expert-gold design has not been executed; it is preserved so that a reader can see what was planned, and it supplies no evidence for anything reported here.

The executed mapping study has its own preregistration chain. The design freezes are `protocol/PUBLIC_REFERENCE_PROTOCOL_V1.json` and `protocol/PUBLIC_REFERENCE_PROTOCOL_V1_1.json`, and the execution freeze for the confirmatory holdout is `protocol/PUBLIC_REFERENCE_CONFIRMATORY_EXECUTION_V1.json`, registered under the identifier `P3.public-reference-mapping.v1.1-confirmatory`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number in Section 9 can be traced to a decision made in advance of seeing it. The successor battery is frozen separately under `research/claim_expansion/p3/`, whose protocol, protected freeze, outcome-access receipt and result are `P3_X_PROTOCOL_V1.md`, `P3_X_PROTECTED_FREEZE_V1.json`, `P3_X_OUTCOME_ACCESS_RECEIPT_V1.md` and `P3_X_PROTECTED_RESULT_V1.json`.

12.2 Mechanism implementation

The coordinate comparison, bridge admissibility and mapping semantics defined in Section 3 are implemented in `src/orion/knowledge/semantics.py`; the public-reference builders, evaluator, analysis and publication projection are `src/orion/study/p3_public_reference_build.py`, `src/orion/study/p3_public_reference_holdout.py`, `src/orion/study/p3_public_reference.py`, `src/orion/study/p3_public_reference_analysis.py` and `src/orion/study/p3_public_reference_publication.py`. The separately implemented repository evaluator used for replay is `src/orion/study/p3_independent_evaluator.py`.

12.3 Frozen public-reference datasets

The initial gold archive is `gold/adjudicated/public-reference-v1/PUBLIC_REFERENCE_GOLD_V1.jsonl` with its freeze manifest `gold/adjudicated/public-reference-v1/PUBLIC_REFERENCE_FREEZE_MANIFEST_V1.json`; its portable SHA-256 is `35f9e39b75ff53b7f0ec82cd03ebcaaa82509ee0aea3f5b96aac3fd62c854ed8`. The disjoint confirmatory holdout is the same pair of files under `gold/adjudicated/public-reference-v1.1-confirmatory/`, with portable SHA-256 `13a76c68c149c2552f3543babeca6e1ad5afe23c45ea9c0dc365c1445cf2782b`. The pinned external source authorities and their revisions are recorded in `gold/PUBLIC_REFERENCE_SOURCE_REGISTRY_V1.json`, and the sampling and authority rules are `gold/PUBLIC_REFERENCE_AUTHORITY_POLICY_V1.md` and `gold/GOLD_METHODOLOGY_V1.md`. Restricted third-party text is not redistributed; the packages retain source identity, version, locator and content commitment instead.

12.4 Frozen results

The initial build report, evaluated summary, analysis, provenance record and checksum manifest are under `evidence/public-reference-v1/`; the external-source revisions and evaluator identity are bound in `evidence/public-reference-v1/PROVENANCE.env`. The confirmatory counterparts are under `evidence/public-reference-v1.1-confirmatory/`: the pre-outcome binding is `EXECUTION_MANIFEST.json`, the reported numbers are `CONFIRMATORY_ANALYSIS.json`, and the publication figures and tables are under `publication/` with their own `SHA256SUMS`. The bootstrap seed 20260817 and both pass margins are recorded in the execution manifest rather than chosen afterwards.

The separate-implementation reproduction of the confirmatory mapping result is `research/verification/records/P3.C5.confirmatory-mapping.json`; the successor battery's separate verifier and its record are `research/claim_expansion/p3/verify_p3_x_independent.py` and `research/claim_expansion/p3/P3_X_INDEPENDENT_VERIFICATION_V1.json`. The bounded method-structure pilot of Section 6.7 has its protocol, runner and results in `research/extensions/p1-p3-structure/` and its receipt at `evidence/method-structure/BOUNDED_PILOT_RECEIPT_V1.md`.

The mapping from each result-bearing claim to the archived artifact that supports it is `CLAIM_LEDGER_V1.md`, with the manuscript map at `CLAIM_LEDGER_MANUSCRIPT_MAP_V1.md`.

12.5 External-successor preflights

The prospective partial-identification design is `protocol/P3_PARTIAL_IDENTIFICATION_SUCCESSOR_V1.json`. Its metadata-only source roots are recorded in `development/naturalistic-panel-harness-2026-08-23/SOURCE_METADATA_SNAPSHOT_V1.json`, whose canonical payload digest is `7dd283acd81e63e84887d82ee7761e89e62a47d31afceca8628e456b76f73e01`; the exact P3 non-promotion status is in development/naturalistic-panel-harness-2026-08-23/PROTOCOL_BINDING_STATUS_V1.json. No raw case, coordinate assignment, gold label or system output occurs in that snapshot. The provider- and modality-diverse successor frame is development/provider-diverse-metadata-census-2026-08-23/SOURCE_CENSUS_V1.json, with protocol development/provider-diverse-metadata-census-2026-08-23/CENSUS_PROTOCOL_V1.json and receipt development/provider-diverse-metadata-census-2026-08-23/VERIFICATION_RECEIPT_V1.json. It closes metadata structure only; its rights, eligibility and scientific-source-universe terminals remain fail-closed.

The comparator-identity audit is development/comparator-binding-harness-2026-08-23/COMPARATOR_BINDING_LEDGER_V1.json. It records that the prospective four-arm multiplicity family is count-only; until four exact identities, revisions, licences, adapters and terminal maps are bound, its terminal remains P3_COMPARATOR_FAMILY_CANNOT_CHECK_IDENTITY.

The public OAEI development protocol, frozen inputs and runners are under development/p3-oaei-public-development-execution-2026-08-23/. The bounded result, failure anatomy and manuscript boundary are RESULT_V2.json, FAILURE_ATLAS_V1.json, RESULTS.md and MANUSCRIPT_READY_BOUNDED_UPDATE.md; their directory manifest is SHA256SUMS. The result digest is 5f5edee08aca44d34e5748d16f1923fba88f3eb47bf2e591c49eea37ee83fdef. Large reproducible intermediate files were hashed and removed rather than committed. This route exposes public references and has authority only as one-family development; it is not independent protected gold.

The distinct post-public-gold-informed cross-construct successor is under development/p3-oaei-cross-construct-successor-2026-08-23/. Its protocol, execution receipts, full result, failure diagnosis, theorem note, coordinate and comparator audits, transport blockers and scientific report are P3_CROSS_CONSTRUCT_PROTOCOL_V3.json, DEVELOPMENT_EXECUTION_FREEZE_RECEIPT_V3.json, RESULT_V3.json, RECURSIVE_FAILURE_DIAGNOSIS_V3.json, IDENTIFICATION_ENVELOPE_THEOREM_V3.md, COORDINATE_IDENTIFICATION_AUDIT_V3.json, COMPARATOR_IDENTITY_AUDIT_V3.json, TRANSPORT_BLOCKERS_V3.json and RESULTS.md. The V3 result SHA-256 is 2c9e59dc323cf65200152ffc259341131e60bb040175f9f109872f35d5d468d4. The large case, prediction and public-gold JSONL products were retained by hash and line count and then removed, as recorded in TEMP_DATA_DISPOSAL_RECEIPT_V3.json. V3 was designed after V2 public gold was opened, so it is public development evidence only and supplies no protected or independent authority.

The distinct outcome-blind V4 feasibility audit is under development/p3-selective-envelope-harm-successor-v4-2026-08-23/. Its protocol, source-rights audit, comparator binding, formal theorem, conditional harm bound, result, adverse-result ledger and verification receipt are P3_SELECTIVE_ENVELOPE_HARM_PROTOCOL_V4.json, SOURCE_FAMILY_RIGHTS.json, COMPARATOR_RESEARCH.json, OPEN_WORLD_REFERENCE_IDENTIFICATION_THEOREM_V4.json, SELECTIVITY_HARM_BOUND_V4.json, RESULT_V4.json, RECURSIVE_NEGATIVE_RESULT_LEDGER_V4.json and AUDIT_RECEIPT_V4.json. All retained bytes are bound by the directory SHA256SUMS; the V4 result SHA-256 is 580bc76eb64d6678a45adb6fac900d8ad18d597548b883bef18471740b50b81b. No new-family reference payload, relation value, matcher output, performance result or protected evidence was opened.

The distinct V5 direct-certificate calibration packet is under development/p3-authoritative-negative-semantics-v5-2026-08-23/. Its freeze, rights/identity audit, 61-file byte manifest, 4,838-row certificate registry, theorem/bound, adverse-result ledger, result and 23-check scientific validation are retained as PROTOCOL_FREEZE_RECEIPT_V5.json, RIGHTS_IDENTITY_AUDIT_V5.json, SOURCE_BYTE_MANIFEST_V5.json, CERTIFICATE_REGISTRY_V5.jsonl, THEOREM_AND_BOUND_V5.json, NEGATIVE_RESULT_LEDGER_V5.json, RESULT_V5.json and

SCIENTIFIC_VALIDATION_V5.json. The V5 result SHA-256 is 5e468fc319fcc70e2f1058912cd34da7290831e2c20801f73e495ee7339cded7. The packet retains no raw ontology payload. No comparator output, protected outcome or naturalistic cross-ontology result was opened.

The V6 comparator-native preflight packet is under development/p3-comparator-native-preflight-v6-2026-08-23/. The three slot results, overall result, negative ledger, runtime manifest, prospective V7 dependency protocol and 63-check validation are K1_AML_RESULT_V6.json, K2_LOGMAP_RESULT_V6.json, K3_BERTMAP_RESULT_V6.json, RESULT_V6.json, NEGATIVE_RESULT_LEDGER_V6.json, RUNTIME_MANIFEST_V6.json, K3_V7_COMPATIBILITY_SUCCESSOR_PROTOCOL.json, and VALIDATION_RECEIPT_V6.json. Their result SHA-256 values are 41821ba01456f674beea1923d95e8ea7cd76641f71050c71ae8ed27d16768139, 44f59b0dbb89545d30a30105b2cfdacc9e31da0bfad637180f0a4d4a6eb3bba1, 89397bde96305b19b19487a87d0b8b82f5270cd8fafcb2993a596c3bd0e64df1, and 9f9a3a8cf8926068ca5dd6a0ff90c586494c18f6087368ce3975d7cc4089a538. Temporary dependencies, model weights, raw native outputs and logs were deleted after bounded hashes/counts were retained. No gold, protected output or comparator score appears in the packet.

The BERTMap compatibility and closed-parser successor is under development/p3-bertmap-execution-binding-v7-2026-08-23/. Its frozen protocol, 26-distribution compatibility lock, runtime/source-rights receipts, five-artifact contract, parser receipt, negative ledger, result and 63-check validation are bound by the local SHA256SUMS. The result SHA-256 is 4190f248bd8f00954d4994f34680ddb29782ad6b946abf38ee235f5b6cb8bf5d. No ontology, model, benchmark, gold, protected output, training, prediction, repair or scoring appears in the packet.

The prospectively frozen table-reader repair is under development/p3-bertmap-table-reader-repair-v8-2026-08-23/. Its protocol and freeze receipt, exact pinned source and Apache-2.0 notice, one-expression patch, language-level execution receipt, result, negative ledger and packet-native validator are bound by the local SHA256SUMS. The patch SHA-256 is 412050c5d3a0e7891a21744a9690d3e4ba06886ea51eec273b6045bff688e42b, and the result SHA-256 is 6ccd4b9ab8dafe8dbf8bcf0c33cb76006ab043c0df34bd5e0d0595ef2d03138b. The validator passes 42/42 checks and the manifest binds 14 files. Temporary five-file parser fixtures were deleted; actual native artifact presence is 0/5. No DeepOnto import, JVM, ontology, model, benchmark, gold, protected output, training, prediction, native mapping repair or scoring appears in this packet.

The complete-runtime V9 native packet is under development/p3-repaired-native-runtime-v9-2026-08-23/. Its prospective protocol, Python/JVM SBOM and rights receipts, single-attempt receipt, four retained native artifacts, execution/parser receipts, cleanup audit, adverse ledger, result and packet-native validator are bound by the local SHA256SUMS. The validator passes 33/33 checks; validation-receipt SHA-256 is 2a6c329587266859de77b1ec164be9055636c8e9b45ab321f3962707be74ffe7. The absent repair artifact and the Java 17/Guice 4.0 exception are retained as fail-closed evidence. Runtime caches and model/source stages were removed after the audit, releasing approximately 5.42 GB; the retained packet is about 9.1 MB. No gold, protected output, correctness, performance, harm, coverage or transport score appears in the packet.

The V10 repair-only stop and V11 repaired runtime-closure successor are under development/p3-java17-add-opens-microgate-v10-2026-08-23/ and development/p3-logmap-manifest-classpath-v11-2026-08-23/. V10 retains the direct missing-OWLAPI failure with 0/90 dependencies. V11 hash-verifies 90/90 manifest dependencies and retains the single exit-zero 16-row repair. Neither packet opens reference truth or runs BERTMap training.

The one-attempt full successor, typed lexical adapter and reference-identity gate are under development/p3-full-native-runtime-v12-2026-08-23/, development/p3-optional-wrapper-typed-decoder-v13-2026-08-23/, and development/p3-public-reference-alignment-v14-2026-08-23/. V12 retains five native artifacts and direct child status. V13 retains the original wrapped repair, twelve adversarial grammar cases and a separately decoded artifact with

SHA-256 c67ee88d541013f41984b239f9cdeaebdcd81573f2080d8af24c7688207dd0f3. V14 retains the 21-unit provider inventory/hash audit and opens no reference file. Its receipt and result SHA-256 digests are b7477e2472b1ab5580beb8191124fe6f3a94c6500642084694c9372f24bf1f8b and fc1f1d35668604e72bc63edd397988eb85c035145b0e6025b848d7c5d3c96314. No protected output, correctness, performance, harm, coverage or transport score appears in V10–V14.

The provider-native identity and exact-Java-8 comparator packets are under development/p3-provider-native-identity-packet-v15-2026-08-23/ and development/p3-aml-java8-prospective-comparator-v16-2026-08-23/. V15 binds the Zenodo/OAEI ontology and reference identities and AML release without semantic reference access; V16 retains the single exit-zero 46-cell AML output. Their SHA256SUMS digests are 1dd1e8234c19e4e53dec4b96d5ecfa05473300a7e44e077dd8de92cc5d5da558 and 8c37aceda665a235088ffa62b0b0c9e718fe75c174184779ea3eeb635461fd96.

The immutable V17 HermiT attempt, V18 structural-surface failure and V19 outcome-blind class-surface successor are under development/p3-oaei-bertmap-common-scoring-v17-2026-08-24/, development/p3-structural-reasoner-compatibility-v18-2026-08-24/ and development/p3-deeponto-class-surface-v19-2026-08-24/. They retain the single-attempt logs, exact adverse terminals, protocol and runtime receipts, universe manifests and packet-native validators. Their SHA256SUMS digests are c1d8bf3c9103b3eb39d15e49b0352552c4c2dcabb21ba8192ad8135c9d38d8ed, eaa0e352a2c4dd6774055e5b5cdd8edf81c55671924cf3f98fb884f016babedd and 46792f76e285a9161950c89fc944b4c37e453c395de7d7ec466549a4b9783168.

The V20 native BERTMap packet and V21 identity-decoder/scoring packet are under development/p3-oaei-bertmap-common-scoring-v20-2026-08-24/ and development/p3-direct-iri-decoder-common-scoring-v21-2026-08-24/. V20 retains the one exit-zero run, all five frozen native-artifact identities and the inherited typed-decoder error without opening the reference. V21 copies and binds those five artifacts, retains the identity-transform decoder and structural-parser receipts, and contains the exact pair table opened only after both matcher outputs were frozen. The metrics and scientific-report SHA-256 digests are 99c2b5614e461587c327a61bffae918b9af4d478efbaacf2f9e2c31135ad3082 and d473b8e4d9858045fa99791ac2fd476744916ecc38103bbf2166e05774531411; the two packet SHA256SUMS digests are 07e4965600e683e90546cb8b53442f7c9927c2e5fce69db2a59c4e5d2e0ec1ff and c6b888f1bdd54baa8eaddc3d9c0f29f4427836096cc9dd79a849e4f82f763baa. These files reproduce one finite public-case description. They do not provide an independent, source-disjoint or population-level comparison.

The authority-composition theory packet is under development/p3-authority-composition-theory-v7t-2026-08-23/. Its formal note, two-world countermodel receipt, result, negative ledger, literature boundary and 32-check validation are AUTHORITY_AUGMENTED_IDENTIFICATION_THEORY_V7T.md, FINITE_COUNTERMODEL_RESULT_V7T.json, RESULT_V7T.json, NEGATIVE_RESULT_LEDGER_V7T.md, NEAREST_WORK_DELTA_V7T.md, and VALIDATION_RECEIPT_V7T.json. No comparator outcome, gold value or protected datum occurs in that packet; it supplies a mathematical composition boundary only.

13 Code availability

Central custom code for the mapping calculus, public-reference builders, evaluators and publication projection is present in this repository snapshot at the paths listed above, with executable instructions in REPRODUCE.md. The current package does not yet bind a public archive URL, archive DOI, immutable submission commit or verified repository-level licence. A submission must therefore give editors and reviewers access to the exact snapshot and bind those four release coordinates before external review; this manuscript does not replace them with an unqualified availability-on-request statement.

13.1 Reproduction

Two make targets regenerate the reported analysis and the publication artifacts from committed inputs: `make paper03-public-reference` produces the headline analysis, and `make paper03-public-reference-publication` produces the publication figures and tables. Both are deterministic, and continuous integration re-runs the second on a clean checkout and requires the regenerated bytes to be identical to the committed ones. The short reproduction route is `REPRODUCE.md`.

Reproduction establishes only what the frozen artifacts establish. The unexecuted raw-text expert atlas, external retrieval and schema baselines, generated-portrait recoverability and downstream answer quality have no artifacts here, and their outcomes remain undetermined rather than absent by oversight.

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